GRADE 8 COMMON CORE MATH

Answer Key

Main • Backward Branches • Forward 1 (Grade 9) • Forward 2 (Grade 10)

65 question sets • 1625 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

CONTENTS

Set 1	**8.NS.A.1**	Classify numbers as rational or irrational	6
Set 2	**8.NS.A.1**	Convert a repeating decimal into a fraction	37
Set 3	**8.NS.A.2**	Estimate the value of an irrational number	68
Set 4	**8.NS.A.2**	Locate and compare irrational numbers on a number line	99
Set 59	**8.NS.A.2**	Estimate the values of expressions that contain irrational numbers	130
Set 5	**8.EE.A.1**	Use the product and quotient properties of integer exponents	161
Set 6	**8.EE.A.1**	Use the power-of-a-power property and zero and negative exponents	192
Set 7	**8.EE.A.2**	Solve equations of the form $x^2 = p$ and $x^3 = p$	223
Set 8	**8.EE.A.2**	Evaluate square roots and cube roots and recognize irrational roots	254
Set 9	**8.EE.A.3**	Estimate very large and very small quantities as a single digit times a power of 10	285
Set 10	**8.EE.A.4**	Multiply and divide numbers written in scientific notation	316
Set 11	**8.EE.A.4**	Add and subtract numbers written in scientific notation	347
Set 12	**8.EE.A.4**	Choose units for scientific notation and read calculator notation	378
Set 13	**8.EE.B.5**	Graph a proportional relationship and interpret the unit rate as the slope	409
Set 14	**8.EE.B.5**	Compare two proportional relationships represented in different ways	440
Set 15	**8.EE.B.6**	Use similar triangles to explain why the slope is the same between any two points	471
Set 60	**8.EE.B.6**	Use similar triangles to derive $y = mx$ for a line through the origin	502

CONTENTS (continued)

Set 61	**8.EE.B.6**	Use similar triangles to derive y = mx + b for a line that crosses the y-axis at b	533
Set 16	**8.EE.C.7.a**	Decide whether a linear equation has one solution, no solution, or infinitely many	564
Set 62	**8.EE.C.7.a**	Give examples of linear equations with one, no, or infinitely many solutions and s	595
Set 17	**8.EE.C.7.b**	Solve multistep linear equations by expanding and collecting like terms	626
Set 18	**8.EE.C.7.b**	Write and solve linear equations with rational coefficients for real-world problems	657
Set 19	**8.EE.C.8.a**	Solve a system of two linear equations by graphing	688
Set 20	**8.EE.C.8.b**	Solve a system of two linear equations algebraically	719
Set 21	**8.EE.C.8.b**	Solve simple systems by inspection	750
Set 22	**8.EE.C.8.c**	Write and solve systems of linear equations for real-world problems	781
Set 23	**8.F.A.1**	Decide whether a relation is a function	812
Set 24	**8.F.A.1**	Graph a function as the set of ordered pairs of inputs and outputs	843
Set 25	**8.F.A.2**	Compare two functions represented in different ways	874
Set 26	**8.F.A.3**	Decide whether a function is linear or nonlinear	905
Set 27	**8.F.B.4**	Find the rate of change and initial value from two points, a table, or a graph	936
Set 28	**8.F.B.4**	Construct a linear function to model a situation	967
Set 29	**8.F.B.4**	Interpret the rate of change and initial value of a linear function in context	998
Set 30	**8.F.B.5**	Describe the relationship shown by a graph	1029

CONTENTS (continued)

Set 31	**8.F.B.5**	Sketch a graph from a description of a relationship	1060
Set 32	**8.G.A.1.a**	Verify that rigid motions take lines to lines and segments to segments of the sam	1091
Set 33	**8.G.A.1.b**	Verify that rigid motions take angles to angles of the same measure	1122
Set 34	**8.G.A.1.c**	Verify that rigid motions take parallel lines to parallel lines	1153
Set 35	**8.G.A.2**	Describe a sequence of rigid motions that shows two figures are congruent	1184
Set 36	**8.G.A.3**	Translate figures on the coordinate plane	1215
Set 37	**8.G.A.3**	Reflect figures across the x-axis and the y-axis	1246
Set 38	**8.G.A.3**	Rotate figures 90° and 180° about the origin	1277
Set 39	**8.G.A.3**	Dilate figures on the coordinate plane with the origin as the center	1308
Set 40	**8.G.A.4**	Describe a sequence of transformations that shows two figures are similar	1339
Set 41	**8.G.A.5**	Use the angle sum and exterior angles of a triangle	1370
Set 63	**8.G.A.5**	Give informal arguments for the triangle angle sum and the exterior angle of a tria	1401
Set 42	**8.G.A.5**	Write and solve equations for angles formed by parallel lines cut by a transversal	1432
Set 64	**8.G.A.5**	Give informal arguments for the angles formed when parallel lines are cut by a tra	1463
Set 43	**8.G.A.5**	Use the angle-angle criterion to show that triangles are similar	1494
Set 65	**8.G.A.5**	Give an informal argument for the angle-angle criterion for similar triangles	1525
Set 44	**8.G.B.6**	Explain a proof of the Pythagorean theorem and its converse	1556

CONTENTS (continued)

Set 45	**8.G.B.6**	Use the converse of the Pythagorean theorem to decide whether a triangle is a ri	1587
Set 46	**8.G.B.7**	Use the Pythagorean theorem to find unknown lengths in two dimensions	1618
Set 47	**8.G.B.7**	Use the Pythagorean theorem to find unknown lengths in three dimensions	1649
Set 48	**8.G.B.8**	Find the distance between two points on the coordinate plane	1680
Set 49	**8.G.C.9**	Find the volume of a cylinder	1711
Set 50	**8.G.C.9**	Find the volume of a cone	1742
Set 51	**8.G.C.9**	Find the volume of a sphere	1773
Set 52	**8.G.C.9**	Solve real-world problems with the volumes of cylinders, cones, and spheres	1804
Set 53	**8.SP.A.1**	Construct a scatter plot for bivariate measurement data	1835
Set 54	**8.SP.A.1**	Describe clustering, outliers, and positive, negative, linear, or nonlinear associati	1866
Set 55	**8.SP.A.2**	Informally fit a straight line to a scatter plot and judge how closely it fits	1897
Set 56	**8.SP.A.3**	Use the equation of a linear model to interpret the slope and intercept and to ma	1928
Set 57	**8.SP.A.4**	Construct a two-way table summarizing data on two categorical variables	1959
Set 58	**8.SP.A.4**	Use relative frequencies in a two-way table to describe a possible association	1990

SET 1 • 8.NS.A.1 • THE NUMBER SYSTEM

Classify numbers as rational or irrational

MAIN	Grade 8 main question	**8.NS.A.1**
BACKWARD 1	Write a fraction as a terminating or repeating decimal	**7.NS.A.2.d**
BACKWARD 2	Write rational numbers as quotients of integers	**7.NS.A.2.b**
FORWARD 1	Decide whether sums and products are rational or irrational	**HSN-RN.B.3**
FORWARD 2	Extend the number system to complex numbers using i	**HSN-CN.A.1**

SET 1 • 8.NS.A.1

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.1

Which number is irrational?

A. $\sqrt{49}$

B. $\frac{-7}{3}$

C. 0.1212... (the digits 12 repeat)

D. $\sqrt{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q1/5

8

QUESTION ID

S1-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. √15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q1/5

Is 0.375 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q2/5

9

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

Rational. It equals 3/8, a quotient of integers (its decimal ends).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q2/5

Is √50 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q3/5

10

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

Irrational. 50 is not a perfect square, so √50 cannot be written as a quotient of integers; its decimal neither ends nor repeats.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q3/5

True or false?

Every number whose decimal expansion repeats is rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q4/5

11

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q4/5

True or false?

π is rational because it is often written as 3.14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q5/5

12

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.14 is only an approximation; π is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 8.NS.A.1

BACKWARD 1

Write a fraction as a terminating or repeating decimal • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{5}{8}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q1/5

14

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.625

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q1/5

Write $\frac{2}{3}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q2/5

15

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.666... (the 6 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q2/5

Which fraction has a repeating decimal?

A. $\frac{3}{5}$

B. $\frac{1}{6}$

C. $\frac{1}{4}$

D. $\frac{7}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q3/5

16

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\frac{4}{11}$ = 0.3636... (the digits 36 repeat)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q4/5

17

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{1}{3} = 0.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q5/5

18

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/3 = 0.333..., a repeating decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 8.NS.A.1

BACKWARD 2

Write rational numbers as quotients of integers • Grade 7 • The Number System • 7.NS.A.2.b

Write $-2\frac{1}{4}$ as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q1/5

20

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−9/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q1/5

Write 0.7 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q2/5

21

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q2/5

Which number is equal to $\frac{-5}{6}$?

A. $\frac{5}{6}$

B. $\frac{-5}{-6}$

C. $\frac{6}{-5}$

D. $\frac{5}{-6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q3/5

22

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5/(−6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q3/5

True or false?

-12 can be written as $\frac{-12}{1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q4/5

23

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\frac{-3}{0}$ is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q5/5

24

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 8.NS.A.1

FORWARD 1

Decide whether sums and products are rational or irrational • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why 4 + √3 is irrational. (Suppose 4 + √3 = r for some rational number r, and see what that would say about √3.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q1/5

26

QUESTION ID

S1-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

4 + √3 is irrational.

GRADING NOTE REQUIRED WORK: An argument like this: if 4 + √3 = r with r rational, then √3 = r − 4, a difference of two rational numbers, which is rational. But √3 is irrational, so 4 + √3 cannot be rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q1**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q1/5

Is √8 × √2 rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q2/5

27

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Rational; √16 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q2/5

Which product is irrational?

A. $\sqrt{3} \times \sqrt{12}$

B. $3\sqrt{5}$

C. $0 \times \sqrt{7}$

D. $\frac{2}{3} \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q3/5

28

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $3\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q3/5

True or false?

The sum of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q4/5

29

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q4/5

True or false?

The product of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q5/5

30

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{2} \times \sqrt{2} = 2$, which is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q5**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 8.NS.A.1

FORWARD 2

Extend the number system to complex numbers using i • High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1

The number i has $i^2 = -1$.

Write $\sqrt{-16}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q1/5

32

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q1/5

The number i has $i^2 = -1$.

What is i^4?

$i^4 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q2/5

33

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q2/5

The number i has $i^2 = -1$.

Which number is a solution of $x^2 = -9$?

A. 3

B. -3

C. 9i

D. 3i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q3/5

34

QUESTION ID

S1-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number a is also a complex number, a + 0i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q4/5

35

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

$\sqrt{-25}$ is a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q5/5

36

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{-25} = 5i$, which is not real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q5/5

SET 2 • 8.NS.A.1 • THE NUMBER SYSTEM

Convert a repeating decimal into a fraction

MAIN	Grade 8 main question	**8.NS.A.1**
BACKWARD 1	Use long division to write a fraction as a repeating decimal	**7.NS.A.2.d**
BACKWARD 2	Multiply decimals by powers of 10	**5.NBT.A.2**
FORWARD 1	Solve linear equations with letter coefficients that come from repe.	**HSA-REI.B.3**
FORWARD 2	Write a geometric sequence from a repeating decimal	**HSF-BF.A.2**

SET 2 • 8.NS.A.1

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 0.444... (the 4 repeats) as a fraction. Show your work.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q1/5

39

QUESTION ID

S2-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

4/9

GRADING NOTE REQUIRED WORK: Any valid method. Example: let x = 0.444...; then 10x = 4.444...; subtracting gives 9x = 4, so x = 4/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q1/5

Write 0.2727... (the digits 27 repeat) as a fraction in simplest form.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q2/5

40

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

3/11

GRADING NOTE 27/99 is equal but is not in simplest form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q2/5

Which fraction is equal to 0.1666... (the 6 repeats)?

A. $\frac{1}{16}$

B. $\frac{1}{6}$

C. $\frac{16}{100}$

D. $\frac{6}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q3/5

41

QUESTION ID

S2-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q3/5

True or false?

2.333... (the 3 repeats) is equal to $\frac{7}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q4/5

42

QUESTION ID

S2-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q4/5

True or false?

0.0909... (the digits 09 repeat) is equal to $\frac{9}{100}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q5/5

43

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is equal to 9/99 = 1/11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 8.NS.A.1

BACKWARD 1

Use long division to write a fraction as a repeating decimal • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{7}{9}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q1/5

45

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.777... (the 7 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q1/5

Write $\frac{5}{12}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q2/5

46

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.41666... (the 6 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{3}{11}$?

A. 0.27 (the decimal ends)

B. 0.3

C. 0.2727... (the digits 27 repeat)

D. 0.311

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q3/5

47

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.2727... (the digits 27 repeat)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{9} = 0.111...$ (the 1 repeats)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q4/5

48

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q4/5

True or false?

$\frac{5}{6} = 0.56$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q5/5

49

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5/6 = 0.8333...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 8.NS.A.1

BACKWARD 2

Multiply decimals by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

0.454 × 100

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q1/5

51

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q1/5

Multiply.

3.27 × 10

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q2/5

52

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

32.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q2/5

Multiply.

0.0081 × 1,000

A. 81

B. 8.1

C. 0.81

D. 0.081

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q3/5

53

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 \times 100 = 250$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q4/5

54

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$0.36 \times 10 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q5/5

55

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 8.NS.A.1

FORWARD 1

Solve linear equations with letter coefficients that come from repeating decimals • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Let x = 0.ddd..., where the digit d (from 1 to 9) repeats. Then 10x = d + x.

Solve for x in terms of d.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q1/5

57

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

d/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q1/5

Let x = 0.ababab..., where the two digits ab repeat, and let n be the two-digit number ab. Then 100x = n + x.

Solve for x in terms of n.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q2/5

58

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

n/99

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q2/5

Solve for x in terms of k. Assume k ≠ 1.

kx = x + 5

A. x = 5 ÷ (k − 1)

B. x = 5 − k

C. x = 5 ÷ k

D. x = 5k − 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q3/5

59

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 5 ÷ (k − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q3/5

True or false?

If $1{,}000x = 123 + x$, then $x = \frac{123}{999} = \frac{41}{333}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q4/5

60

QUESTION ID

S2-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q4/5

True or false?

For every number m, the equation mx = 2x + 3 has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q5/5

61

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE When m = 2, it becomes 0 = 3, which has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 8.NS.A.1

FORWARD 2

Write a geometric sequence from a repeating decimal • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2

0.333... = 0.3 + 0.03 + 0.003 + ... The terms 0.3, 0.03, 0.003, ... form a geometric sequence.

What is the common ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q1/5

63

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.1

GRADING NOTE 1/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q1/5

Write an explicit formula for the nth term of the sequence 0.3, 0.03, 0.003, ...

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q2/5

64

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.3(0.1)^(n − 1)

GRADING NOTE Equivalent forms such as 3(0.1)^n are correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q2/5

Which recursive formula gives the sequence 0.27, 0.0027, 0.000027, ...?

A. $a(1) = 0.27;\ a(n) = 100 \cdot a(n-1)$

B. $a(1) = 0.27;\ a(n) = 0.01 \cdot a(n-1)$

C. $a(1) = 0.27;\ a(n) = a(n-1) - 0.01$

D. $a(1) = 0.0027;\ a(n) = 0.01 \cdot a(n-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q3/5

65

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $a(1) = 0.27;\ a(n) = 0.01 \cdot a(n-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q3/5

True or false?

The sequence 0.5, 0.05, 0.005, ... is geometric with a common ratio of $\frac{1}{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q4/5

66

QUESTION ID

S2-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

The sequence 0.4, 0.04, 0.004, ... is arithmetic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q5/5

67

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The terms have a common ratio, not a common difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • 8.NS.A.2 • THE NUMBER SYSTEM

Estimate the value of an irrational number

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Find squares of whole numbers and decimals	**6.EE.A.1**
BACKWARD 2	Round decimals to the nearest tenth	**5.NBT.A.4**
FORWARD 1	Solve quadratic equations by taking square roots, exactly and ap…	**HSA-REI.B.4.b**
FORWARD 2	Find arc lengths and sector areas exactly in terms of π and appro…	**HSG-C.B.5**

SET 3 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Between which two whole numbers is √40?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q1/5

70

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

6 and 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q1/5

Estimate √20 to the nearest tenth. Explain how you know.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q2/5

71

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

4.5. $4.4^2 = 19.36$ and $4.5^2 = 20.25$, and 20 is closer to 20.25.

GRADING NOTE Grade both: 4.5, and a reason that compares squares of tenths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q2/5

Which is the best estimate of √75 to the nearest tenth?

A. 8.7

B. 8.6

C. 37.5

D. 9.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q3/5

72

QUESTION ID

S3-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q3/5

True or false?

√2 is between 1.4 and 1.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q4/5

73

QUESTION ID

S3-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q4/5

True or false?

√90 is between 8 and 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q5/5

74

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $9^2 = 81$ and $10^2 = 100$, so √90 is between 9 and 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q5/5

SET 3 • 8.NS.A.2

BACKWARD 1

Find squares of whole numbers and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

7^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q1/5

76

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q1/5

Find the value.

1.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q2/5

77

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q2/5

Find the value.

12^2

A. 24

B. 121

C. 122

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q3/5

78

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$8^2 = 64$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q4/5

79

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

$3^2 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q5/5

80

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 8.NS.A.2

BACKWARD 2

Round decimals to the nearest tenth • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 4.472 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q1/5

82

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q1/5

Round 8.660 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q2/5

83

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q2/5

Round 3.162 to the nearest tenth.

A. 3.1

B. 3.0

C. 3.2

D. 3.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q3/5

84

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q3/5

True or false?

6.325 rounded to the nearest tenth is 6.3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q4/5

85

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q4/5

True or false?

2.449 rounded to the nearest tenth is 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q5/5

86

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 8.NS.A.2

FORWARD 1

Solve quadratic equations by taking square roots, exactly and approximately • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 = 30$. Give the exact solutions, then round each to the nearest tenth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q1/5

88

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \sqrt{30}$ and $x = -\sqrt{30}$; about 5.5 and −5.5

GRADING NOTE Both solutions are required, exact and rounded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q1/5

Solve $x^2 - 7 = 0$. Give the exact solutions.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q2/5

89

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = \sqrt{7}$ and $x = -\sqrt{7}$

GRADING NOTE Both solutions are required; $x = \pm\sqrt{7}$ is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q2/5

Solve $(x - 1)^2 = 5$.

A. $x = 1 + \sqrt{5}$ or $x = 1 - \sqrt{5}$

B. $x = 1 + \sqrt{5}$ only

C. $x = 6$ or $x = -4$

D. $x = \sqrt{6}$ or $x = -\sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q3/5

90

QUESTION ID

S3-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $x = 1 + \sqrt{5}$ or $x = 1 - \sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q3/5

True or false?

The solutions of $x^2 = 12$ are $x = 2\sqrt{3}$ and $x = -2\sqrt{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q4/5

91

QUESTION ID

S3-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q4/5

True or false?

$x^2 = -9$ has the real solutions 3 and −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q5/5

92

QUESTION ID

S3-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No real number squared is negative, so there is no real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q5/5

SET 3 • 8.NS.A.2

FORWARD 2

Find arc lengths and sector areas exactly in terms of π and approximately • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

A circle has a radius of 6 cm.

Find the length of a 90° arc exactly in terms of π, then to the nearest tenth.

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q1/5 • NEAREST RELATED

94

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3π cm, about 9.4 cm

GRADING NOTE Both the exact and the rounded value are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q1/5 • NEAREST RELATED

A circle has a radius of 4 inches.

Find the area of a 45° sector exactly in terms of π, then to the nearest tenth.

Sector area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q2/5 • NEAREST RELATED

95

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2π square inches, about 6.3 square inches

GRADING NOTE Both the exact and the rounded value are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q2/5 • NEAREST RELATED

A circle has a radius of 3 m.

To the nearest tenth, how long is a 60° arc?

A. 6.3 m

B. 9.4 m

C. 3.1 m

D. 18.8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q3/5 • NEAREST RELATED

96

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3.1 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

The arc of a semicircle with a radius of 1 is π units long, about 3.14 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q4/5 • NEAREST RELATED

97

QUESTION ID

S3-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A 180° sector of a circle with a radius of 2 has an area of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q5/5 • NEAREST RELATED

98

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is half of 4π, which is 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 4 • 8.NS.A.2 • THE NUMBER SYSTEM

Locate and compare irrational numbers on a number line

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Read inequalities as positions on a number line	**6.NS.C.7.a**
BACKWARD 2	Position decimals on a number line	**6.NS.C.6.c**
FORWARD 1	Report irrational values to a level of accuracy that fits the situati…	**HSN-Q.A.3**
FORWARD 2	Use coordinates to find exact lengths and perimeters with radicals	**HSG-GPE.B.7**

SET 4 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Plot √5 on the number line. Use an estimate to the nearest tenth.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q1/5

101

QUESTION ID

S4-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A point at about 2.2: between 2 and 2.5, closer to 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q1/5

Which list is in order from least to greatest?

A. 2, √5, √3, 1.8

B. √3, 1.8, 2, √5

C. √3, 2, 1.8, √5

D. 1.8, √3, 2, √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q2/5

102

QUESTION ID

S4-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. √3, 1.8, 2, √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q2/5

Write <, >, or =.

π ___ √10

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q3/5

103

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

<

GRADING NOTE π is about 3.14 and √10 is about 3.16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q3/5

Which number is the best name for point P: √2, √3, or √6?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q4/5

104

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q4/5

True or false?

√30 is to the left of 5.5 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q5/5

105

QUESTION ID

S4-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE √30 is about 5.48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q5/5

SET 4 • 8.NS.A.2

BACKWARD 1

Read inequalities as positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

2.4 is to the right of 2.25 on a number line.

Write an inequality that shows this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q1/5

107

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.4 > 2.25

GRADING NOTE 2.25 < 2.4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q1/5

Which number is farthest to the left on a number line?

A. 0.5

B. −1.9

C. −2.05

D. −2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q2/5

108

QUESTION ID

S4-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. −2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q2/5

What does 1.7 < 1.75 tell you about the positions of the two numbers on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q3/5

109

QUESTION ID

S4-B1-Q3

TYPE

Short answer

CORRECT ANSWER

1.7 is to the left of 1.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q3/5

True or false?

$-1.4 > -1.41$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q4/5

110

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q4/5

True or false?

3.09 > 3.1

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q5/5

111

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 8.NS.A.2

BACKWARD 2

Position decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

Plot 1.4 and 2.75 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q1/5

113

QUESTION ID

S4-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

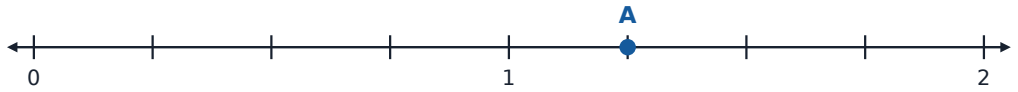
Points at 1.4 (a little past the 1.25 mark) and 2.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q1/5

What decimal is at point A?



Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q2/5

114

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q2/5

Which decimal is between 1.7 and 1.8?

A. 1.81

B. 1.73

C. 1.69

D. 1.08

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q3/5

115

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.73

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q3/5

True or false?

2.5 is halfway between 2 and 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q4/5

116

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q4/5

True or false?

0.9 is to the right of 1.1 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q5/5

117

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 8.NS.A.2

FORWARD 1

Report irrational values to a level of accuracy that fits the situation • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

A square garden has an area of 50 square meters, so its side is √50 m. A tape measure reads to the nearest centimeter.

Report the side length to that accuracy, in meters.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q1/5

119

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7.07 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q1/5

A coin has a diameter of 4 cm. Its circumference is 4π cm.

Report the circumference to the nearest tenth of a centimeter.

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q2/5

120

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q2/5

Which value of √2 is given to the nearest thousandth?

A. 1.41

B. 1.414

C. 1.4142

D. 1.42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q3/5

121

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.414

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q3/5

True or false?

Reporting √3 as 1.7 is accurate to the nearest tenth.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q4/5

122

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q4/5

True or false?

Writing more digits of √5 always makes a measured length more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q5/5

123

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A reported length cannot be more accurate than the measuring tool.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 8.NS.A.2

FORWARD 2

Use coordinates to find exact lengths and perimeters with radicals • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the exact perimeter of the triangle with vertices (0, 0), (2, 0), and (0, 2).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q1/5

125

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 2√2 units

GRADING NOTE The exact form is required; a decimal approximation alone is not accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q1/5

Find the exact length of the segment from (1, 1) to (4, 3).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q2/5

126

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

√13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q2/5

A square has vertices (0, 0), (1, 2), (−1, 3), and (−2, 1).

What is its perimeter?

A. 2√5 units

B. 20 units

C. 4√5 units

D. 4√3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q3/5

127

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4√5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q3/5

True or false?

The distance from $(-1, -1)$ to $(2, 3)$ is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q4/5

128

QUESTION ID

S4-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q4/5

True or false?

A rectangle with vertices (0, 0), (3, 0), (3, 1), and (0, 1) has a diagonal 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q5/5

129

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The diagonal is √10 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q5/5

SET 59 • 8.NS.A.2 • THE NUMBER SYSTEM

Estimate the values of expressions that contain irrational numbers

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Evaluate expressions with decimal values	**6.EE.A.2.c**
BACKWARD 2	Add, subtract, and multiply multi-digit decimals	**6.NS.B.3**
FORWARD 1	Solve quadratic equations and estimate their irrational solutions	**HSA-REI.B.4.b**
FORWARD 2	Derive the quadratic formula by completing the square	**HSA-REI.B.4.a**

SET 59 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Estimate 2√5 + 1 to the nearest tenth. Show how you estimated √5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q1/5

132

QUESTION ID

S59-M-Q1

TYPE

Short answer

CORRECT ANSWER

About 5.5. √5 ≈ 2.24 ($2.2^2 = 4.84$ and $2.3^2 = 5.29$; $2.24^2 \approx 5.02$), so $2\sqrt{5} + 1 \approx 4.48 + 1 = 5.48 \approx 5.5$.

GRADING NOTE Grade both: 5.5, and an estimate of √5 between 2.2 and 2.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q1/5

Estimate π^2 to the nearest whole number.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q2/5

133

QUESTION ID

S59-M-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE $\pi \approx 3.14$, and $3.14^2 \approx 9.86$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q2/5

Which is the best estimate of √2 + √3 to the nearest tenth?

A. 5.0

B. 3.1

C. 3.6

D. 2.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q3/5

134

QUESTION ID

S59-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q3/5

Between which two whole numbers is $3\sqrt{7}$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q4/5

135

QUESTION ID

S59-M-Q4

TYPE

Short answer

CORRECT ANSWER

7 and 8. $3\sqrt{7} = \sqrt{63}$, and $7^2 = 49 < 63 < 64 = 8^2$.

GRADING NOTE Any correct reasoning is accepted, for example $\sqrt{7} \approx 2.65$ so $3\sqrt{7} \approx 7.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q4/5

True or false?

$5 - \sqrt{2}$ is greater than 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q5/5

136

QUESTION ID

S59-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{2} \approx 1.41$, so $5 - \sqrt{2} \approx 3.59$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q5/5

SET 59 • 8.NS.A.2

BACKWARD 1

Evaluate expressions with decimal values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 2x + 1 when x = 2.24.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q1/5

138

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5.48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q1/5

Evaluate x^2 when $x = 3.14$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q2/5

139

QUESTION ID

S59-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9.8596

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q2/5

Evaluate 3x − 1 when x = 1.7.

A. 3.1

B. 4.1

C. 5.1

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q3/5

140

QUESTION ID

S59-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q3/5

True or false?

When x = 1.4, x + 1.7 = 3.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q4/5

141

QUESTION ID

S59-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q4/5

True or false?

When x = 2.6, 3x = 6.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q5/5

142

QUESTION ID

S59-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 × 2.6 = 7.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q5/5

SET 59 • 8.NS.A.2

BACKWARD 2

Add, subtract, and multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

1.414 + 1.732

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q1/5

144

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.146

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q1/5

Multiply.

2×2.236

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q2/5

145

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4.472

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q2/5

Subtract.

$3.162 - 1$

A. 2.062

B. 4.162

C. 3.062

D. 2.162

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q3/5

146

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2.162

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q3/5

True or false?

$9.86 + 0.14 = 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q4/5

147

QUESTION ID

S59-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q4/5

True or false?

$3 \times 2.65 = 7.65$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q5/5

148

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3 \times 2.65 = 7.95$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • 8.NS.A.2

FORWARD 1

Solve quadratic equations and estimate their irrational solutions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 - 4x + 1 = 0$. Give the exact solutions, then estimate each to the nearest tenth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q1/5

150

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 2 + \sqrt{3} \approx 3.7$ and $x = 2 - \sqrt{3} \approx 0.3$

GRADING NOTE Both solutions are required, exact and estimated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q1/5

Solve $(x + 1)^2 = 8$. Give the exact solutions, then estimate each to the nearest tenth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q2/5

151

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = -1 + 2\sqrt{2} \approx 1.8$ and $x = -1 - 2\sqrt{2} \approx -3.8$

GRADING NOTE Both solutions are required, exact and estimated. $-1 \pm \sqrt{8}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q2/5

Which is the best estimate of the positive solution of $x^2 + 2x - 4 = 0$?

A. 1.2

B. −3.2

C. 2.2

D. 3.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q3/5

152

QUESTION ID

S59-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q3/5

True or false?

The solutions of $x^2 = 2x + 2$ are $1 + \sqrt{3}$ and $1 - \sqrt{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q4/5

153

QUESTION ID

S59-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q4/5

True or false?

The positive solution of $x^2 = 6$ is about 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q5/5

154

QUESTION ID

S59-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{6} \approx 2.4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q5/5

SET 59 • 8.NS.A.2

FORWARD 2

Derive the quadratic formula by completing the square • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

Complete the square to solve $x^2 + 6x + c = 0$ for x in terms of c. Assume $c \leq 9$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q1/5

156

QUESTION ID

S59-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x = -3 + \sqrt{(9 - c)}$

and

$x = -3 - \sqrt{(9 - c)}$

GRADING NOTE REQUIRED WORK: The completed square must be shown: $x^2 + 6x + 9 = 9 - c$, so $(x + 3)^2 = 9 - c$; then $x + 3 = \pm\sqrt{(9 - c)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q1/5

Start with $ax^2 + bx + c = 0$ ($a \neq 0$). Divide by a, subtract c/a, and add $(b/(2a))^2$ to both sides.

What equation do you get?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q2/5

157

QUESTION ID

S59-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + b/(2a))^2 = (b^2 - 4ac)/(4a^2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q2**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q2/5

Which is the quadratic formula for $ax^2 + bx + c = 0$ ($a \neq 0$)?

A. $x = (b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$

B. $x = (-b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$

C. $x = (-b \pm \sqrt{(b^2 + 4ac)}) \div (2a)$

D. $x = -b \pm \sqrt{(b^2 - 4ac)} \div 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q3/5

158

QUESTION ID

S59-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $x = (-b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q3/5

True or false?

For $x^2 - 2x - 1 = 0$, the quadratic formula gives $x = 1 \pm \sqrt{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q4/5

159

QUESTION ID

S59-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q4/5

True or false?

Completing the square on $x^2 + 10x = 3$ gives $(x + 5)^2 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q5/5

160

QUESTION ID

S59-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives $(x + 5)^2 = 28$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q5/5

SET 5 • 8.EE.A.1 • EXPRESSIONS & EQUATIONS

Use the product and quotient properties of integer exponents

MAIN	Grade 8 main question	**8.EE.A.1**
BACKWARD 1	Evaluate whole-number exponents	**6.EE.A.1**
BACKWARD 2	Write powers of 10 with whole-number exponents	**5.NBT.A.2**
FORWARD 1	Multiply and divide monomials using exponent properties	**HSA-SSE.A.2**
FORWARD 2	Use exponent properties to rewrite and interpret exponential funct	**HSF-IF.C.8.b**

SET 5 • 8.EE.A.1

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$3^4 \times 3^5$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q1/5

163

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

3^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q1/5

Write as a single power.

$7^{10} \div 7^{4}$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q2/5

164

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

7^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q2/5

Which expression is equal to 2^6?

A. $2^{12} \div 2^2$

B. 2×6

C. $2^2 \times 2^3$

D. $2^2 \times 2^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q3/5

QUESTION ID

S5-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $2^2 \times 2^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q3/5

True or false?

$5^3 \times 5^2 = 5^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q4/5

166

QUESTION ID

S5-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q4/5

True or false?

$4^8 \div 4^2 = 4^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q5/5

167

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4^8 \div 4^2 = 4^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q5/5

SET 5 • 8.EE.A.1

BACKWARD 1

Evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

2^5

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q1/5

169

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

10^4

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q2/5

170

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q2/5

Find the value.

3^4

A. 81

B. 64

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q3/5

171

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$6^2 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q4/5

172

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q4/5

True or false?

$2^3 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q5/5

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 8.EE.A.1

BACKWARD 2

Write powers of 10 with whole-number exponents • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Write 10 × 10 × 10 × 10 as a power of 10.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q1/5

175

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10^4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q1/5

Write 1,000,000 as a power of 10.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q2/5

176

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q2/5

What is the value of 10^5?

A. 10,000

B. 50

C. 100,000

D. 1,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q3/5

177

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 100,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q3/5

True or false?

$10^3 = 1{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q4/5

178

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q4/5

True or false?

$10^2 \times 10 = 10^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q5/5

179

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10^2 \times 10 = 1{,}000 = 10^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 8.EE.A.1

FORWARD 1

Multiply and divide monomials using exponent properties • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Simplify.

$(3x^4)(5x^2)$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q1/5

181

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$15x^6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q1/5

Simplify. Assume $x \neq 0$.

$$\frac{12x^7}{4x^3}$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q2/5

182

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$3x^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q2/5

Simplify.

$x^5 \cdot x \cdot x^2$

A. x^7

B. $3x^8$

C. x^{10}

D. x^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q3/5

183

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q3/5

True or false?

$(2a^3)(4a^2) = 8a^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q4/5

184

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q4/5

True or false?

Assume $y \neq 0$. $\frac{y^9}{y^3} = y^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q5/5

185

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $y^9 \div y^3 = y^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 8.EE.A.1

FORWARD 2

Use exponent properties to rewrite and interpret exponential functions • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b

f(t) = 2^(t + 3)

Use a property of exponents to write f(t) in the form a · 2^t.

f(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q1/5

187

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8 · 2^t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q1/5

A population is P(t) = 200 · 2^t after t years.

By what factor does the population grow each year, and what is P(5)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q2/5

188

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It doubles (factor 2) each year;
P(5) = 6,400

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q2/5

$g(t) = 5^{(t - 1)}$

Which expression is equal to g(t)?

A. $5^t - 5$

B. $5 \div 5^t$

C. $5^t \div 5$

D. $5^t - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q3/5

189

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $5^t \div 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q3/5

True or false?

If f(x) = 5 · 2^x, then f(x + 1) = 2 · f(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q4/5

190

QUESTION ID

S5-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q4/5

True or false?

If f(x) = 2^x, then f(3) · f(4) = f(12).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q5/5

191

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(3) · f(4) = $2^3 \cdot 2^4 = 2^7$ = f(7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q5/5

SET 6 • 8.EE.A.1 • EXPRESSIONS & EQUATIONS

Use the power-of-a-power property and zero and negative exponents

MAIN	Grade 8 main question	**8.EE.A.1**
BACKWARD 1	Find powers of fractions and decimals	**6.EE.A.1**
BACKWARD 2	Multiply signed numbers	**7.NS.A.2.a**
FORWARD 1	Explain why rational exponents are defined so the properties of ex…	**HSN-RN.A.1**
FORWARD 2	Rewrite exponential expressions with the properties of exponents	**HSA-SSE.B.3.c**

SET 6 • 8.EE.A.1

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$(5^2)^3$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q1/5

194

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

5^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q1/5

Find the value.

4^{-2}

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q2/5

195

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q2/5

Which expression is equal to 1?

A. 0^9

B. 9^0

C. $1^{-9} \times 9$

D. 9^{-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q3/5

196

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9^0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q3/5

True or false?

$3^{-2} \times 3^{5} = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q4/5

197

QUESTION ID

S6-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q4/5

True or false?

$(2^3)^2 = 2^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q5/5

198

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(2^3)^2 = 2^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q5/5

SET 6 • 8.EE.A.1

BACKWARD 1

Find powers of fractions and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

$(\frac{1}{2})^3$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q1/5

200

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

0.3^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q2/5

201

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.09

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q2/5

Find the value.

$(\frac{2}{3})^2$

A. $\frac{4}{3}$

B. $\frac{4}{9}$

C. $\frac{2}{9}$

D. $\frac{4}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q3/5

202

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$0.1^3 = 0.001$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q4/5

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q4/5

True or false?

$(\frac{3}{4})^2 = \frac{6}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q5/5

204

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(3/4)^2 = 9/16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 8.EE.A.1

BACKWARD 2

Multiply signed numbers • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

$(-2)(-2)(-2)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q1/5

206

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q1/5

Multiply.

$(-3)(-3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q2/5

207

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q2/5

Multiply.

$(-1)(-1)(-1)(-1)$

A. -1

B. -4

C. 1

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q3/5

208

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q3/5

True or false?

$(-5)(-5) = 25$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q4/5

209

QUESTION ID

S6-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q4/5

True or false?

$(-2)(-2)(-2)(-2) = -16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q5/5

210

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 8.EE.A.1

FORWARD 1

Explain why rational exponents are defined so the properties of exponents still hold • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1

If the power-of-a-power property is to stay true for rational exponents, then $(5^{1/2})^2 = 5^{(1/2) \cdot 2} = 5$.

Explain what number $5^{1/2}$ must be.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q1/5

212

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$5^{1/2}$ must be a positive number whose square is 5, so $5^{1/2} = \sqrt{5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q1/5

If the power-of-a-power property is to stay true, what must 8^(1/3) equal? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q2/5

213

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2, because $(8^{1/3})^3 = 8^{(1/3)\cdot 3} = 8$, and 2 is the number whose cube is 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q2/5

Which number is equal to 9^(3/2)?

A. 6

B. 27

C. 13.5

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q3/5

214

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q3/5

True or false?

Because $(8^{(1/3)})^3 = 8$, $8^{(1/3)} = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q4/5

215

QUESTION ID

S6-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q4/5

True or false?

x^(1/2) means x ÷ 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q5/5

216

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x^(1/2) is the square root of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q5**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 8.EE.A.1

FORWARD 2

Rewrite exponential expressions with the properties of exponents • High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c

Rewrite 1.21^t as a power of 1.1.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q1/5

218

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.1^(2t)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q1/5

A value grows by a factor of 1.04 each month, so after t years it has grown by (1.04^12)^t.

Write the yearly growth factor, rounded to the nearest hundredth.

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q2/5

219

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.60

GRADING NOTE 1.04^12 ≈ 1.601, about 60% growth per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q2/5

Which expression is equivalent to 8^x?

A. 2^x ÷ 3

B. 3^(2x)

C. 2^(3x)

D. 2^(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q3/5

220

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2^(3x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q3/5

True or false?

$(0.5)^{(-t)} = 2^t$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q4/5

221

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q4/5

True or false?

$4^x \times 4^x = 16^{(2x)}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q5/5

222

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4^x \times 4^x = 4^{(2x)} = 16^x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q5/5

SET 7 • 8.EE.A.2 • EXPRESSIONS & EQUATIONS

Solve equations of the form $x^2 = p$ and $x^3 = p$

MAIN	Grade 8 main question	**8.EE.A.2**
BACKWARD 1	Solve one-step multiplication equations	**6.EE.B.7**
BACKWARD 2	Find squares and cubes of whole numbers	**6.EE.A.1**
FORWARD 1	Solve quadratic equations by taking square roots	**HSA-REI.B.4.b**
FORWARD 2	Find a dimension from a volume with square and cube roots	**HSG-GMD.A.3**

SET 7 • 8.EE.A.2

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x. Give all solutions.

$x^2 = 49$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q1/5

225

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7 and x = −7

GRADING NOTE Both solutions are required; x = ±7 is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q1/5

Solve for x.

$x^3 = 64$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q2/5

226

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q2/5

Solve for x.

$x^2 = \frac{9}{16}$

A. $x = \frac{3}{4}$ only

B. $x = \frac{3}{4}$ or $x = -\frac{3}{4}$

C. $x = \frac{9}{32}$

D. $x = \frac{81}{256}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q3/5

227

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 3/4 or x = −3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q3/5

Solve for x.

$x^3 = -27$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q4/5

228

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q4/5

True or false?

$x^2 = 10$ has the two solutions $x = \sqrt{10}$ and $x = -\sqrt{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q5/5

229

QUESTION ID

S7-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q5/5

SET 7 • 8.EE.A.2

BACKWARD 1

Solve one-step multiplication equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

4x = 64

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q1/5

231

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q1/5

Solve for x.

$\frac{x}{3} = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q2/5

232

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q2/5

Solve for x.

9x = 45

A. x = 36

B. x = 5

C. x = 405

D. x = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q3/5

233

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q3/5

True or false?

x = 12 is the solution of 3x = 36.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q4/5

234

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q4/5

True or false?

x = 5 is the solution of 7x = 42.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q5/5

235

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 8.EE.A.2

BACKWARD 2

Find squares and cubes of whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

9^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q1/5

237

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q1/5

Find the value.

5^3

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q2/5

238

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which whole number cubed is 8?

A. 3

B. 4

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q3/5

239

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q3/5

True or false?

$10^4 = 10{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q4/5

240

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q4/5

True or false?

$4^2 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q5/5

241

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 8.EE.A.2

FORWARD 1

Solve quadratic equations by taking square roots • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve for x. Give all solutions.

$2x^2 = 50$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q1/5

243

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 5 and x = −5

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q1/5

Solve for x. Give all solutions.

$(x + 2)^2 = 36$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q2/5

244

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4 and x = −8

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q2/5

Solve for x.

$x^2 - 9 = 0$

A. x = 9 or x = −9

B. x = 3 only

C. x = 3 or x = −3

D. x = 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q3/5

245

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 3 or x = −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q3/5

True or false?

The solutions of $3x^2 = 27$ are 3 and −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q4/5

246

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q4/5

True or false?

The solutions of $(x - 1)^2 = 16$ are 5 and 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q5/5

247

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solutions are 5 and −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q5/5

SET 7 • 8.EE.A.2

FORWARD 2

Find a dimension from a volume with square and cube roots • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Every edge of a cube is multiplied by the same factor, and the volume is multiplied by 27.

By what factor was each edge multiplied? Explain.

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q1/5

249

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3, because the volume is multiplied by the cube of the factor and $\sqrt[3]{27} = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q1/5

A sphere has a volume of 36π cubic centimeters.

($V = \frac{4}{3}\pi r^3$)

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q2/5

250

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q2/5

A cylinder has a volume of 200π cubic units and a height of 8 units.

What is its radius?

A. 10 units

B. 25 units

C. 5 units

D. √200 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q3/5

251

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q3/5

True or false?

A cone with a volume of 12π cubic units and a height of 4 units has a radius of 3 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q4/5

252

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q4/5

True or false?

A cube with a volume of 1,000 cubic centimeters has edges 100 cm long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q5/5

253

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each edge is 10 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q5/5

SET 8 • 8.EE.A.2 • EXPRESSIONS & EQUATIONS

Evaluate square roots and cube roots and recognize irrational roots

MAIN	Grade 8 main question	**8.EE.A.2**
BACKWARD 1	Find factor pairs and square numbers	**4.OA.B.4**
BACKWARD 2	Find the volume of a cube	**5.MD.C.5.b**
FORWARD 1	Simplify square roots by factoring out perfect squares	**HSN-RN.A.2**
FORWARD 2	Solve quadratic equations by completing the square	**HSA-REI.B.4.a**

SET 8 • 8.EE.A.2

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.2

Find the value.

√144

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q1/5

256

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q1/5

Find the value.

$\sqrt[3]{125}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q2/5

257

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q2/5

Which square root is irrational?

A. $\sqrt{\frac{1}{4}}$

B. $\sqrt{0.09}$

C. $\sqrt{20}$

D. $\sqrt{36}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q3/5

258

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. √20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q3/5

True or false?

$\sqrt[3]{(-8)} = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q4/5

259

QUESTION ID

S8-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q4/5

True or false?

√2 can be written as a fraction of two whole numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q5/5

260

QUESTION ID
S8-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE √2 is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q5/5

SET 8 • 8.EE.A.2

BACKWARD 1

Find factor pairs and square numbers • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 16.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q1/5

262

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 × 16, 2 × 8, and 4 × 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q1/5

What whole number times itself is 49?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q2/5

263

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q2/5

Which number has a factor pair made of two equal factors?

A. 30

B. 24

C. 18

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q3/5

264

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q3/5

True or false?

81 has a factor pair made of two equal factors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q4/5

265

QUESTION ID

S8-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q4/5

True or false?

12 has a factor pair made of two equal factors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q5/5

266

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 8.EE.A.2

BACKWARD 2

Find the volume of a cube • Grade 5 • Measurement & Data • 5.MD.C.5.b

A cube has edges 3 units long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q1/5

268

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

27 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q1/5

A cube has edges 5 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q2/5

269

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

125 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q2/5

Which cube has a volume of 64 cubic units?

A. A cube with 32-unit edges

B. A cube with 4-unit edges

C. A cube with 16-unit edges

D. A cube with 8-unit edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q3/5

270

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A cube with 4-unit edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q3/5

True or false?

A cube with 2-unit edges has a volume of 8 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q4/5

271

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q4/5

True or false?

A cube with 10-unit edges has a volume of 30 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q5/5

272

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 8.EE.A.2

FORWARD 1

Simplify square roots by factoring out perfect squares • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify.

√48

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q1/5

274

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q1**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q1/5

Simplify.

√200

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q2/5

275

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q2**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q2/5

Which expression is equal to √18?

A. 2√3

B. 9√2

C. 3√2

D. 6√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q3/5

276

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q3/5

True or false?

$\sqrt{50} = 5\sqrt{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q4/5

277

QUESTION ID

S8-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q4/5

True or false?

$\sqrt{27} = 9\sqrt{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q5/5

278

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{27} = 3\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q5/5

SET 8 • 8.EE.A.2

FORWARD 2

Solve quadratic equations by completing the square • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

What number goes in the blank?

$x^2 + 6x + 9 = (x + ___)^2$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q1/5

280

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q1**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q1/5

Solve by completing the square. Show the completed square, then give both exact solutions.

$x^2 + 4x = 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q2/5

281

QUESTION ID

S8-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

$x = -2 + \sqrt{5}$ and $x = -2 - \sqrt{5}$

GRADING NOTE REQUIRED WORK: The completed square must be shown: $x^2 + 4x + 4 = 1 + 4$, so $(x + 2)^2 = 5$; then $x + 2 = \pm\sqrt{5}$. Both roots are required. Correct roots without the completed square are not full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q2/5

Which number completes the square?

$x^2 - 10x + ___$

A. 5

B. −25

C. 100

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q3/5

282

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q3/5

True or false?

$x^2 + 8x + 16 = (x + 4)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q4/5

283

QUESTION ID

S8-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q4/5

True or false?

Completing the square on $x^2 + 2x = 3$ gives $(x + 1)^2 = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q5/5

284

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives $(x + 1)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q5/5

SET 9 • 8.EE.A.3 • EXPRESSIONS & EQUATIONS

Estimate very large and very small quantities as a single digit times a power of 10

MAIN	Grade 8 main question	**8.EE.A.3**
BACKWARD 1	Multiply and divide by powers of 10	**5.NBT.A.2**
BACKWARD 2	Round whole numbers to any place	**4.NBT.A.3**
FORWARD 1	Choose a level of accuracy for very large and very small quantities	**HSN-Q.A.3**
FORWARD 2	Use base-10 logarithms to solve exponential equations	**HSF-LE.A.4**

SET 9 • 8.EE.A.3

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.3

Estimate 412,000,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q1/5

287

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

4×10^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q1/5

Estimate 0.00079 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q2/5

288

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

8×10^{-4}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q2/5

A country has about 3×10^8 people. A city has about 6×10^5 people.

About how many times as many people live in the country?

A. 2

B. 50

C. 500

D. 5,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q3/5

289

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q3/5

A whale has a mass of about 2×10^{5} kg. A mouse has a mass of about 2×10^{-2} kg.

About how many times as large is the whale's mass?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q4/5

290

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

10,000,000

GRADING NOTE 10^{7} is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q4/5

True or false?

6×10^{-3} is greater than 9×10^{-4}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q5/5

291

QUESTION ID

S9-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q5/5

SET 9 • 8.EE.A.3

BACKWARD 1

Multiply and divide by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

4×10^3

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q1/5

293

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q1/5

Divide.

$7 \div 10^2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q2/5

294

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.07

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q2/5

What is the value of 3×10^5?

A. 15

B. 3,000,000

C. 300,000

D. 30,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q3/5

295

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 300,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$6 \div 10^3 = 0.006$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q4/5

296

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$8 \times 10^2 = 8{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q5/5

297

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 8.EE.A.3

BACKWARD 2

Round whole numbers to any place • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3

Round 412,000 to the nearest hundred thousand.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q1/5

299

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

400,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q1/5

Round 6,750 to the nearest thousand.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q2/5

300

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q2/5

Round 58,300 to the nearest ten thousand.

A. 50,000

B. 59,000

C. 60,000

D. 58,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q3/5

301

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 60,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

245,000 rounded to the nearest hundred thousand is 200,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q4/5

302

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

8,500 rounded to the nearest thousand is 8,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q5/5

303

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 8.EE.A.3

FORWARD 1

Choose a level of accuracy for very large and very small quantities • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

Earth's mean radius is about 6,371 km.

Write it to one significant digit as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q1/5

305

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6×10^3 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q1/5

A scale reads to the nearest 0.1 kg. One bag of rice measures 2.3 kg, and a truck carries 4,000 such bags.

Report the total mass in scientific notation, with no more accuracy than the scale supports.

Total mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q2/5

306

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About 9.2 × 10³ kg

GRADING NOTE 9.2 × 10³ kg (two significant digits). Reports such as 9.200 × 10³ kg claim more accuracy than the scale gives.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q2/5

A runner's time over 100 m was measured to the nearest second as 12 s. A calculator gives $100 \div 12 = 8.333333$.

Which report of the speed fits the accuracy of the measurement?

A. About 8 m/s

B. 83 m/s

C. 8.33 m/s

D. 8.333333 m/s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q3/5

QUESTION ID

S9-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. About 8 m/s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reporting 7,900,000,000 people as 8×10^9 is accurate to one significant digit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q4/5

308

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q4/5

True or false?

A distance known to be about 3×10^8 m is reported more accurately as 300,000,000.0 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q5/5

309

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Writing more digits does not make an estimate more accurate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 8.EE.A.3

FORWARD 2

Use base-10 logarithms to solve exponential equations • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

A bacteria count is 10^t thousand after t hours.

Use a logarithm to find when the count reaches 1,000,000 (1,000 thousand).

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q1/5

311

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3 hours

($t = \log_{10}(1{,}000) = 3$)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q1/5

Solve 5 · 10^x = 2,000. Write x as a logarithm, then use a calculator to round it to the nearest tenth.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q2/5

312

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x = \log_{10}(400) \approx 2.6$

GRADING NOTE Both the logarithm and the rounded value are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q2**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q2/5

Between which two whole numbers is $\log_{10}(4{,}500)$?

A. 2 and 3

B. 45 and 46

C. 4 and 5

D. 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q3/5

313

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q3**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q3/5

True or false?

$\log_{10}(10^{-5}) = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q4/5

314

QUESTION ID

S9-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q4/5

True or false?

$\log_{10}(50)$ is between 5 and 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q5/5

315

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10^1 = 10$ and $10^2 = 100$, so $\log_{10}(50)$ is between 1 and 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q5/5

SET 10 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Multiply and divide numbers written in scientific notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Multiply decimals	**5.NBT.B.7**
BACKWARD 2	Multiply by powers of 10	**5.NBT.A.2**
FORWARD 1	Use units to guide multistep calculations in scientific notation	**HSN-Q.A.1**
FORWARD 2	Model exponential growth and decay with scientific notation	**HSF-LE.A.1.c**

SET 10 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

Multiply. Write the answer in scientific notation.

$(4 \times 10^{6})(2.5 \times 10^{3})$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q1/5

318

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

1×10^{10}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q1/5

Divide. Write the answer in scientific notation.

$(6.3 \times 10^{8}) \div (9 \times 10^{2})$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q2/5

319

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

7×10^{5}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q2/5

Light travels about 3×10^5 km per second.

How far does it travel in 5×10^2 seconds?

A. 1.5×10^7 km

B. 6×10^2 km

C. 1.5×10^8 km

D. 8×10^7 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q3/5

320

QUESTION ID

S10-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.5×10^8 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q3/5

True or false?

$(2 \times 10^{-3})(3 \times 10^{-4}) = 6 \times 10^{-7}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q4/5

321

QUESTION ID

S10-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q4/5

True or false?

$(8 \times 10^9) \div (4 \times 10^3) = 2 \times 10^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q5/5

322

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient is 2×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q5/5

SET 10 • 8.EE.A.4

BACKWARD 1

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

2.5 × 4

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q1/5

324

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q1/5

Multiply.

1.2 × 3.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q2/5

325

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q2/5

Multiply.

6×0.5

A. 6.5

B. 3

C. 0.3

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q3/5

326

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$2.4 \times 0.5 = 1.2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q4/5

327

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$3.5 \times 2 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q5/5

328

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 8.EE.A.4

BACKWARD 2

Multiply by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

2.5×10^4

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q1/5

330

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

25,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q1/5

Multiply.

3.1×10^2

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q2/5

331

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

310

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q2/5

Multiply.

0.6×10^3

A. 60

B. 0.0006

C. 6,000

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q3/5

332

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$7 \times 10^3 = 7{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q4/5

333

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q4/5

True or false?

$4.2 \times 10 = 4.20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q5/5

334

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 8.EE.A.4

FORWARD 1

Use units to guide multistep calculations in scientific notation • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

A drive transfers 2×10^8 bytes per second.

How many seconds does it take to transfer 6×10^{10} bytes?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q1/5

336

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q1/5

Light travels 3×10^8 meters per second.

How many meters does it travel in 1 hour (3.6×10^3 seconds)? Use scientific notation.

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q2/5

337

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.08×10^{12} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q2**

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q2/5

The Sun is about 1.5×10^{11} m from Earth. Light travels 3×10^{8} m per second.

About how long does sunlight take to reach Earth?

A. 500 seconds

B. 4.5×10^{19} seconds

C. 5×10^{19} seconds

D. 50 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q3/5

338

QUESTION ID

S10-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 500 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q3/5

True or false?

A flow of 5×10^3 liters per minute is 3×10^5 liters per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q4/5

339

QUESTION ID

S10-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q4/5

True or false?

6×10^{-3} km is equal to 6×10^{-6} m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q5/5

340

QUESTION ID

S10-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6×10^{-3} km = 6 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 8.EE.A.4

FORWARD 2

Model exponential growth and decay with scientific notation • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A colony of 2×10^3 bacteria triples every hour.

How many bacteria are there after 4 hours? Use scientific notation.

Bacteria: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q1/5

342

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.62×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q1**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q1/5

A quantity of 5×10^9 is cut in half each year.

What is it after 3 years? Use scientific notation.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q2/5

343

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6.25×10^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q2**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q2/5

A population of 4×10^6 grows by 10% each year.

What is the population after 2 years?

A. 4.84×10^6

B. 8×10^6

C. 4.4×10^6

D. 4.8×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q3/5

344

QUESTION ID

S10-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4.84×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q3/5

True or false?

Doubling 3×10^5 ten times gives about 3.07×10^8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q4/5

345

QUESTION ID

S10-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q4/5

True or false?

A quantity that grows by a constant percent rate increases by the same amount each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q5/5

346

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q5/5

SET 11 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Add and subtract numbers written in scientific notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Add decimals	**5.NBT.B.7**
BACKWARD 2	Write a number times a power of 10 in standard form	**5.NBT.A.2**
FORWARD 1	Add and subtract polynomials	**HSA-APR.A.1**
FORWARD 2	Add and subtract functions — nearest related (no direct Grade 10+)	**HSF-BF.A.1.b**

SET 11 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

Add. Write the answer in scientific notation.

$(3.2 \times 10^5) + (4.5 \times 10^4)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q1/5

349

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

3.65×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q1/5

Subtract. Write the answer in scientific notation.

$(9 \times 10^{7}) - (6 \times 10^{6})$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q2/5

350

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

8.4×10^{7}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q2/5

Add. Write the answer in scientific notation.

$(1.5 \times 10^{-3}) + (2 \times 10^{-4})$

A. 1.7×10^{-4}

B. 3.5×10^{-3}

C. 1.7×10^{-3}

D. 3.5×10^{-7}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q3/5

351

QUESTION ID

S11-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.7×10^{-3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q3/5

True or false?

$(5 \times 10^8) + (5 \times 10^8) = 1 \times 10^9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q4/5

352

QUESTION ID

S11-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q4/5

True or false?

$(4 \times 10^6) - (3 \times 10^5) = 1 \times 10^1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q5/5

353

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 3.7×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q5/5

SET 11 • 8.EE.A.4

BACKWARD 1

Add decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Add.

3.2 + 0.45

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q1/5

355

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q1/5

Add.

8.4 + 1.6

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q2/5

356

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q2/5

Add.

1.5 + 0.2

A. 0.17

B. 1.7

C. 1.52

D. 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q3/5

357

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q3/5

True or false?

$2.75 + 0.25 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q4/5

358

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q4/5

True or false?

4.6 + 0.04 = 5

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q5/5

359

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 8.EE.A.4

BACKWARD 2

Write a number times a power of 10 in standard form • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Write 6×10^6 in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q1/5

361

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q1/5

Write 0.6×10^7 in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q2/5

362

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q2/5

Which number is equal to 45×10^4?

A. 450,000

B. 4,500

C. 45,000

D. 4,500,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q3/5

363

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 450,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q3/5

True or false?

$3.2 \times 10^2 = 320$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q4/5

364

QUESTION ID

S11-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

$0.5 \times 10^3 = 50$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q5/5

365

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 8.EE.A.4

FORWARD 1

Add and subtract polynomials • High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Add.

$(3x^2 + 5x) + (4x^2 - 2x)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q1/5

367

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$7x^2 + 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q1/5

Subtract.

$(6a^3 - 2a + 1) - (2a^3 + 5a - 4)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q2/5

368

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$4a^3 - 7a + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q2/5

Add.

$(x^2 - 4) + (3x^2 + x)$

A. $4x^2 + x - 4$

B. $3x^4 + x - 4$

C. $4x^2 - 3$

D. $4x^2 + x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q3/5

369

QUESTION ID

S11-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $4x^2 + x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q3/5

True or false?

$(2y^2 + 3) - (y^2 - 3) = y^2 + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q4/5

370

QUESTION ID

S11-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q4/5

True or false?

$(5n - 2) - (3n - 4) = 2n - 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q5/5

371

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 2n + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q5/5

SET 11 • 8.EE.A.4

FORWARD 2

Add and subtract functions • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

Revenue is R(x) = 25x and cost is C(x) = 4,000 + 9x for x items.

Write the profit function P(x) = R(x) − C(x).

P(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q1/5 • NEAREST RELATED

373

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

16x − 4,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q1/5 • NEAREST RELATED

The profit function is P(x) = 16x − 4,000.

Find P(1,000).

P(1,000) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q2/5 • NEAREST RELATED

374

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q2/5 • NEAREST RELATED

$f(x) = 3x^2$ and $g(x) = x^2 - 5$.

Which expression is $(f + g)(x)$?

A. $2x^2 + 5$

B. $4x^2 - 5$

C. $3x^4 - 5$

D. $4x^2 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q3/5 • NEAREST RELATED

375

QUESTION ID

S11-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4x^2 - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

If f(x) = 2x and g(x) = x + 1, then (f − g)(3) = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q4/5 • NEAREST RELATED

376

QUESTION ID

S11-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If $f(x) = x + 4$ and $g(x) = 2x$, then $(f + g)(x) = 2x^2 + 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q5/5 • NEAREST RELATED

377

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(f + g)(x) = 3x + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 12 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Choose units for scientific notation and read calculator notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Convert metric units	**5.MD.A.1**
BACKWARD 2	Divide by powers of 10	**5.NBT.A.2**
FORWARD 1	Convert units of very large and very small quantities	**HSN-Q.A.1**
FORWARD 2	Define the quantities and units a model needs	**HSN-Q.A.2**

SET 12 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

A calculator shows 4.7E8.

Write the number in scientific notation and in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q1/5

380

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

4.7×10^8

$= 470{,}000{,}000$

GRADING NOTE Grade both forms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q1/5

A calculator shows 2.05E-6.

Write the number in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q2/5

381

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.00000205

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q2/5

The distance from Earth to the Moon is about 3.84×10^8 m.

Which unit is the most sensible for this distance?

A. Kilometers

B. Centimeters

C. Millimeters

D. Micrometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q3/5

382

QUESTION ID

S12-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q3/5

A dust particle has a mass of 7.53×10^{-10} kg.

Write the mass in micrograms. (1 μg = 10^{-9} kg)

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q4/5

383

QUESTION ID

S12-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.753 μg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q4/5

True or false?

A calculator display of 3E-4 means −30,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q5/5

384

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3E-4 means $3 \times 10^{-4} = 0.0003$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q5/5

SET 12 • 8.EE.A.4

BACKWARD 1

Convert metric units • Grade 5 • Measurement & Data • 5.MD.A.1

How many grams are in 3.5 kilograms?

Grams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

386

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

How many meters are in 4,200 millimeters?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

387

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

How many centimeters are in 6 meters?

A. 60

B. 0.06

C. 6,000

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

388

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

True or false?

5 km = 5,000 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

389

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

True or false?

250 cm = 25 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

390

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 8.EE.A.4

BACKWARD 2

Divide by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Divide.

$3{,}840 \div 10^3$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

392

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

Divide.

$75 \div 10^2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

393

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

Divide.

$6 \div 10^4$

A. 0.006

B. 0.00006

C. 60,000

D. 0.0006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

394

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.0006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

True or false?

520 ÷ 10 = 52

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

395

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

True or false?

$9 \div 10^3 = 0.09$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

396

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 8.EE.A.4

FORWARD 1

Convert units of very large and very small quantities • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

Convert 3.6×10^6 mm to kilometers.

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q1/5

398

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3.6 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q1/5

A server sends 4.5×10^9 bytes per minute.

Write this rate in gigabytes per second. (1 GB = 10^9 bytes)

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q2/5

399

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.075 GB per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q2/5

A virus is 1.2×10^{-7} m wide.

What is its width in nanometers? (1 nm = 10^{-9} m)

A. 1,200 nm

B. 120 nm

C. 12 nm

D. 0.12 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q3/5

400

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 120 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q3/5

True or false?

7×10^5 g = 700 kg

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q4/5

401

QUESTION ID

S12-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q4/5

True or false?

2.5×10^{-3} km = 2.5 cm

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q5/5

402

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.5×10^{-3} km = 2.5 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q5/5

SET 12 • 8.EE.A.4

FORWARD 2

Define the quantities and units a model needs • High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2

You want to estimate how much water a city of 2×10^6 people uses each day.

Which quantities do you need, and with what units?

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q1/5

404

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The number of people and the water used per person per day (liters per person per day).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q1/5

A model for the cost of running a fan is $C = p \cdot w \cdot h$, with C in dollars.

Define p, w, and h, with units, so that the units work out to dollars.

Definitions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q2/5

405

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

p = the price of electricity (dollars per kilowatt-hour), w = the fan's power (kilowatts), h = the time it runs (hours).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q2/5

A model will predict how long it takes to fill a pool with a hose.

Which pair of quantities does it need?

A. The pool's volume (liters) and the hose's flow rate (liters per minute)

B. The hose's length (meters) and its color

C. The number of swimmers and the pool's depth (meters)

D. The pool's depth (meters) and the water's temperature (°C)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q3/5

406

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The pool's volume (liters) and the hose's flow rate (liters per minute)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q3/5

True or false?

In the model $C = p \cdot w \cdot h$, if p is in dollars per kilowatt-hour, w in kilowatts, and h in hours, then C is in dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q4/5

QUESTION ID

S12-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q4/5

True or false?

A model of a city's daily water use needs only the city's population; the units of the other quantities do not matter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q5/5

408

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It also needs the water used per person per day, with its units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q5/5

SET 13 • 8.EE.B.5 • EXPRESSIONS & EQUATIONS

Graph a proportional relationship and interpret the unit rate as the slope

MAIN	Grade 8 main question	**8.EE.B.5**
BACKWARD 1	Explain points on the graph of a proportional relationship	**7.RP.A.2.d**
BACKWARD 2	Find the constant of proportionality from a graph	**7.RP.A.2.b**
FORWARD 1	Calculate and interpret an average rate of change	**HSF-IF.B.6**
FORWARD 2	Compare properties of functions represented in different ways	**HSF-IF.C.9**

SET 13 • 8.EE.B.5

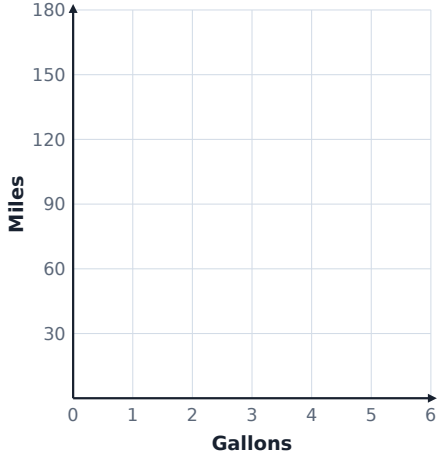
MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.5

A car uses gas at a constant rate: 3 gallons for every 90 miles.

Graph the relationship between gallons (x) and miles (y). What is the slope, and what does it mean?

Slope and meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q1/5

411

QUESTION ID

S13-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line through (0, 0) and points such as (1, 30) and (3, 90)

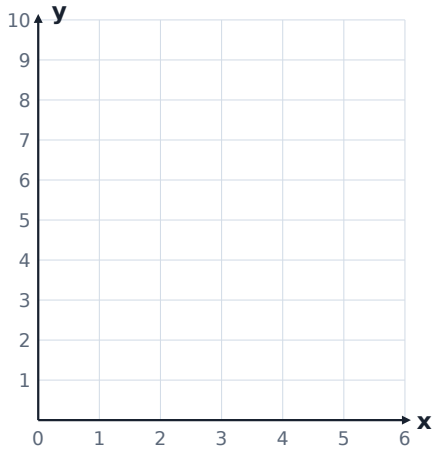
Written: 30; the car travels 30 miles per gallon.

GRADING NOTE Grade both: the line through the origin with slope 30, and the slope with its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q1/5

Graph y = 1.5x on the grid for x from 0 to 6.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q2/5

412

QUESTION ID

S13-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A straight line through (0, 0), (2, 3), (4, 6), and (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

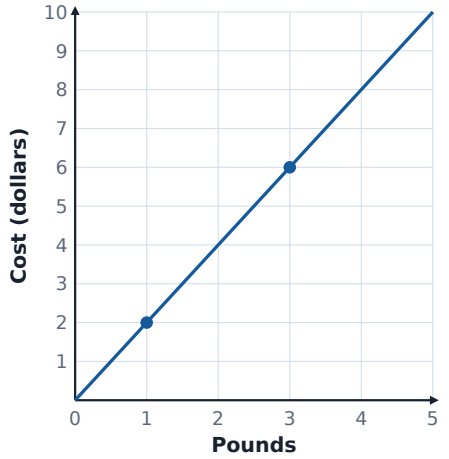
ID S13-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q2/5

The graph shows the cost of apples.

What is the slope, and what does it represent?

Slope and meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q3/5

413

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

2; the apples cost $2 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q3/5

The graph of a proportional relationship passes through (4, 10).

What is the unit rate?

A. 14

B. 2.5

C. 0.4

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q4/5

414

QUESTION ID

S13-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q4/5

True or false?

On graphs of distance (y) against time (x) that pass through the origin, a steeper line means a greater speed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q5/5

415

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q5/5

SET 13 • 8.EE.B.5

BACKWARD 1

Explain points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

A proportional graph of hours (x) and pay in dollars (y) passes through (1, 15).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q1/5

417

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The pay is $15 for 1 hour (the unit rate).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q1/5

A proportional graph shows cups of flour (x) and cookies (y).

What does the point (0, 0) mean?

A. Each cup makes 0 cookies.

B. 0 cups of flour make 0 cookies.

C. The recipe starts with some cookies.

D. There is no graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q2/5

418

QUESTION ID

S13-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 0 cups of flour make 0 cookies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q2/5

A proportional graph of minutes (x) and pages read (y) passes through (5, 20).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q3/5

419

QUESTION ID

S13-B1-Q3

TYPE

Short answer

CORRECT ANSWER

20 pages are read in 5 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q3/5

True or false?

On the graph of a proportional relationship, the point (1, r) tells you the unit rate is r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q4/5

420

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q4/5

True or false?

On a graph of hours (x) and miles (y), the point (3, 12) means 3 miles in 12 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q5/5

421

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 12 miles in 3 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 8.EE.B.5

BACKWARD 2

Find the constant of proportionality from a graph • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A proportional graph passes through (0, 0) and (5, 20).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q1/5

423

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q1/5

Which point shows k = 3 on the graph of a proportional relationship?

A. (3, 6)

B. (6, 2)

C. (3, 1)

D. (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q2/5

424

QUESTION ID

S13-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q2/5

A proportional graph passes through (8, 2).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q3/5

425

QUESTION ID

S13-B2-Q3

TYPE

Short answer

CORRECT ANSWER

0.25

GRADING NOTE 1/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q3/5

True or false?

A line through (0, 0) and (10, 25) has a constant of proportionality of 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q4/5

426

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q4/5

True or false?

A line through (0, 0) and (6, 3) has a constant of proportionality of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q5/5

427

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $k = 3 \div 6 = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 8.EE.B.5

FORWARD 1

Calculate and interpret an average rate of change • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = 3x + 2$

What is the average rate of change of f from $x = 1$ to $x = 5$?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q1/5

429

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q1/5

The table shows a trip.

What is the average rate of change from 2 hours to 5 hours?

Hours	0	2	5
Miles	0	110	260

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q2/5

430

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q2/5

$h(t) = t^2$

What is the average rate of change of h from t = 1 to t = 4?

A. 5

B. 3

C. 16

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q3/5

431

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q3/5

True or false?

For a linear function, the average rate of change is the same over every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q4/5

QUESTION ID

S13-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q4/5

True or false?

For $f(x) = x^2$, the average rate of change from $x = 0$ to $x = 2$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q5/5

433

QUESTION ID
S13-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is $(4 - 0) \div 2 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 8.EE.B.5

FORWARD 2

Compare properties of functions represented in different ways • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = 2.5x$. The table shows $g(x) = x^2$.

Which function has the greater average rate of change from $x = 0$ to $x = 4$? Give both rates.

x	0	1	2	3	4
g(x)	0	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q1/5

435

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

g: its average rate is 16 ÷ 4 = 4, and f's is 2.5.

GRADING NOTE Both rates and the comparison are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q1/5

f(x) = 40 + 5x and g(x) = 30(1.2)^x.

Which function has the greater initial value? Which is greater at x = 10? Round values to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q2/5

436

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

f has the greater initial value (40 > 30); g is greater at x = 10 (about 185.8, while f(10) = 90).

GRADING NOTE Both comparisons are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q2/5

f(x) = 2^x and g(x) = 10x.

For which x-value is f(x) > g(x)?

A. x = 3

B. x = 4

C. x = 6

D. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q3/5

437

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q3/5

True or false?

A linear function with a slope of 4 eventually grows more slowly than f(x) = 2^x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q4/5

438

QUESTION ID

S13-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q4/5

True or false?

Two functions with the same y-intercept must be the same function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q5/5

439

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q5/5

SET 14 • 8.EE.B.5 • EXPRESSIONS & EQUATIONS

Compare two proportional relationships represented in different ways

MAIN	Grade 8 main question	**8.EE.B.5**
BACKWARD 1	Write an equation for a proportional relationship	**7.RP.A.2.c**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Create linear and quadratic equations in two variables and graph t	**HSA-CED.A.2**
FORWARD 2	Compare linear and exponential growth	**HSF-LE.A.3**

SET 14 • 8.EE.B.5

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.5

Runner A's distance is d = 6.2t (miles, hours). Runner B's graph is a line through (0, 0) and (2, 13).

Which runner is faster?

A. Runner A

B. Runner B

C. They run at the same speed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q1/5

442

QUESTION ID

S14-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Runner B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q1/5

Plan A costs c = 0.12m dollars for m minutes. The table shows Plan B.

Which plan costs less per minute, and by how much?

Minutes	100	250	400
Cost ($)	15	37.5	60

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q2/5

443

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

Plan A, by $0.03 per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q2/5

Pump A fills 45 gallons in 3 minutes at a constant rate.
Pump B fills according to g = 14m.

Which pump is faster, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q3/5

444

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

Pump A, by 1 gallon per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q3/5

True or false?

The graph of y = 0.8x is less steep than the graph of a proportional relationship through (5, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q4/5

445

QUESTION ID

S14-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q4/5

True or false?

A proportional relationship through (3, 7) has a greater unit rate than y = 2.5x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q5/5

446

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7 ÷ 3 ≈ 2.33, which is less than 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q5/5

SET 14 • 8.EE.B.5

BACKWARD 1

Write an equation for a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Pencils cost $0.45 each.

Write an equation for the cost c of n pencils.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q1/5

448

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 0.45n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	2	4	6
y	7	14	21

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q2/5

449

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3.5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q2/5

A car travels 55 miles per hour.

Which equation gives the distance d after t hours?

A. $d = \frac{t}{55}$

B. $d = 55t$

C. $t = 55d$

D. $d = t + 55$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q3/5

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. d = 55t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q3/5

True or false?

A recipe uses 6 cups of water for every cup of rice. The equation y = 6x gives the cups of water y for x cups of rice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q4/5

451

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q4/5

True or false?

A printer prints 30 pages in 2 minutes. The equation p = 30m gives the pages p printed in m minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q5/5

452

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is p = 15m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 8.EE.B.5

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ty walks $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

How many miles per hour is this?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q1/5

454

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q1/5

$\frac{2}{5}$ pound of nuts costs $2.

What is the price per pound?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q2/5

455

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q2/5

A hose fills $\frac{1}{2}$ gallon in $\frac{1}{6}$ minute.

How many gallons per minute is this?

A. $\frac{1}{12}$

B. 6

C. $\frac{1}{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q3/5

456

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

Walking $\frac{1}{3}$ mile in $\frac{1}{6}$ hour is a speed of 2 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q4/5

457

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q4/5

True or false?

A drink uses $\frac{3}{4}$ cup of juice for every $\frac{1}{2}$ cup of water. That is $\frac{3}{8}$ cup of juice per cup of water.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q5/5

458

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3/2 cups of juice per cup of water.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 8.EE.B.5

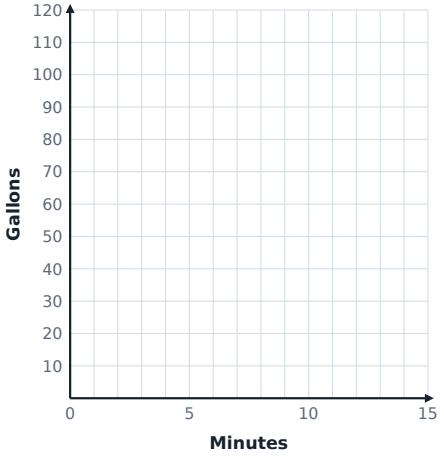
FORWARD 1

Create linear and quadratic equations in two variables and graph them with labels and scales • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

Tank A has 20 gallons and gains 5 gallons per minute. Tank B starts empty and gains 7 gallons per minute.

Write an equation for each tank (y gallons after x minutes). Graph both for $0 \le x \le 15$, label each line, and mark where they meet.

Equations: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q1/5

460

QUESTION ID

S14-F1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Line A from (0, 20) to (15, 95) and line B from (0, 0) to (15, 105), each labeled, meeting at (10, 70)

Written: Tank A: y = 5x + 20; Tank B: y = 7x

GRADING NOTE Grade both: the two equations, and the two labeled lines meeting at (10, 70).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q1/5

A rectangle has a perimeter of 40 m.

Write an equation for its area A in terms of its width w, and give the values of w that make sense.

Equation and domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q2/5

461

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A = w(20 − w), for 0 < w < 20

GRADING NOTE Equivalent forms such as $A = 20w - w^2$ are correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q2/5

A ball's height is $h = -5t^2 + 20t$ meters after t seconds, for $0 \le t \le 4$.

Which axes and scales fit a graph of this equation?

A. Horizontal: time from 0 to 4 seconds; vertical: height from −20 to 0 meters

B. Horizontal: time t from 0 to 4 seconds; vertical: height h from 0 to 20 meters

C. Horizontal: height from 0 to 4 meters; vertical: time from 0 to 20 seconds

D. Horizontal: time from 0 to 20 seconds; vertical: height from 0 to 4 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q3/5

462

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Horizontal: time t from 0 to 4 seconds; vertical: height h from 0 to 20 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q3/5

True or false?

The graphs of y = 5x + 20 and y = 7x intersect at (10, 70).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q4/5

463

QUESTION ID

S14-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q4/5

True or false?

For a rectangle with a perimeter of 40 m, the area $A = w(20 - w)$ is greatest when $w = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q5/5

464

QUESTION ID

S14-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is greatest when w = 10 (a 10 m by 10 m square).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q5/5

SET 14 • 8.EE.B.5

FORWARD 2

Compare linear and exponential growth • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 100 + 50x and g(x) = 100(1.5)^x.

Which is greater at x = 5? Give both values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q1/5

466

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

g: g(5) = 759.375, while f(5) = 350

GRADING NOTE Exact values are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q1/5

Which function eventually has the greater values:
y = 1,000x or y = 2^x?

A. It depends only on the y-intercepts

B. y = 1,000x

C. y = 2^x

D. Neither; they are always equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q2/5

467

QUESTION ID

S14-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q2/5

Which function in the table is exponential? Explain.

x	0	1	2	3
f(x)	2	4	8	16
g(x)	2	4	6	8

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q3/5

468

QUESTION ID

S14-F2-Q3

TYPE

Short answer

CORRECT ANSWER

f, because it doubles each time x increases by 1 (g increases by 2 each time).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q3/5

True or false?

A quantity that increases exponentially eventually exceeds a quantity that increases linearly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q4/5

469

QUESTION ID

S14-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q4/5

True or false?

At x = 2, 3^x is greater than 10x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q5/5

470

QUESTION ID

S14-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^2 = 9$, which is less than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q5/5

SET 15 • 8.EE.B.6 • EXPRESSIONS & EQUATIONS

Use similar triangles to explain why the slope is the same between any two points on a line

MAIN	Grade 8 main question	**8.EE.B.6**
BACKWARD 1	Use a scale factor to find a length in a scaled copy	**7.G.A.1**
BACKWARD 2	Write an equation from a table of values	**6.EE.C.9**
FORWARD 1	Use similarity transformations to decide whether triangles are sim	**HSG-SRT.A.2**
FORWARD 2	Use slopes of parallel and perpendicular lines	**HSG-GPE.B.5**

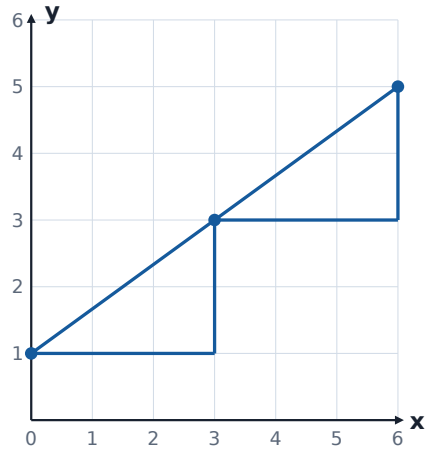
SET 15 • 8.EE.B.6

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.6

Two slope triangles are drawn under the line. Explain why they give the same slope.

Explanation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q1/5

473

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

The triangles are similar (they have the same angles), so their rise-to-run ratios are equal: 2/3 = 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q1/5

A slope triangle on a line has a run of 4 and a rise of 6.
A larger slope triangle on the same line has a run of 10.

Use similar triangles to find its rise.

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q2/5

474

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

15 (6/4 = rise/10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q2/5

Why are two slope triangles on the same non-vertical line similar?

A. Their hypotenuses are always equal.

B. Each has a right angle, and their angles where the line meets the horizontal legs are equal (corresponding angles).

C. Their legs are always equal.

D. They always have equal areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q3/5

475

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Each has a right angle, and their angles where the line meets the horizontal legs are equal (corresponding angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q3/5

The points (2, 1), (5, 3), and (11, 7) lie on one line.

Find the slope using (2, 1) and (5, 3), and again using (5, 3) and (11, 7). Explain why the two results must agree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q4/5

476

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

2/3 and 4/6 = 2/3. The two slope triangles are similar, so rise ÷ run is the same.

GRADING NOTE Grade both slopes and the similar-triangle reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q4/5

True or false?

Any two slope triangles drawn on the same non-vertical line are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q5/5

477

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q5/5

SET 15 • 8.EE.B.6

BACKWARD 1

Use a scale factor to find a length in a scaled copy • Grade 7 • Geometry • 7.G.A.1

A right triangle has legs of 2 and 3. In a scaled copy, the leg of 2 becomes 8.

How long is the other leg in the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q1/5

479

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q1/5

A right triangle has legs of 4 and 6. A scaled copy has a leg of 6 matching the leg of 4.

How long is the other leg in the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q2/5

480

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q2/5

Which pair of right triangles are scaled copies of each other (matching the shorter legs)?

A. Legs 2 and 5; legs 4 and 7

B. Legs 2 and 5; legs 6 and 12

C. Legs 2 and 5; legs 3 and 6

D. Legs 2 and 5; legs 4 and 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q3/5

481

QUESTION ID

S15-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Legs 2 and 5; legs 4 and 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q3/5

True or false?

A scaled copy with a scale factor of 3 has every length 3 times as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q4/5

482

QUESTION ID

S15-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q4/5

True or false?

Right triangles with legs 3 and 4 and legs 6 and 9 are scaled copies of each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q5/5

483

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 8.EE.B.6

BACKWARD 2

Write an equation from a table of values • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x.

x	1	2	3
y	6	7	8

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q1/5

485

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q1/5

Write an equation for y in terms of x.

x	1	2	3
y	3	6	9

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q2/5

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q2/5

Which equation matches the table?

x	0	1	2
y	1	3	5

A. y = 2x

B. y = x + 1

C. y = 3x

D. y = 2x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q3/5

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 2x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q3/5

True or false?

For y = 2x + 4, y = 10 when x = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q4/5

488

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q4/5

True or false?

For y = x + 7, y = 14 when x = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q5/5

489

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 8.EE.B.6

FORWARD 1

Use similarity transformations to decide whether triangles are similar • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides 3, 4, and 5. Triangle DEF has sides 6, 8, and 10.

Are the triangles similar? If so, what is the scale factor from ABC to DEF?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q1/5

491

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; the scale factor is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q1/5

On the line through (0, 1), (3, 3), and (9, 7), slope triangle T1 has vertices (0, 1), (3, 1), (3, 3). Slope triangle T2 has vertices (3, 3), (9, 3), (9, 7).

Describe a translation followed by a dilation that maps T1 onto T2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q2/5

492

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Translate T1 3 units right and 2 units up, so (0, 1) → (3, 3), (3, 1) → (6, 3), (3, 3) → (6, 5). Then dilate about (3, 3) with a scale factor of 2: (6, 3) → (9, 3) and (6, 5) → (9, 7).

GRADING NOTE Any correct sequence that names the translation, the center, and the scale factor is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q2/5

Which information is enough to show that two triangles are similar?

A. One pair of equal sides

B. Two pairs of equal angles

C. Equal perimeters

D. One pair of equal angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q3/5

493

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Two pairs of equal angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q3/5

True or false?

If one triangle is a dilation of another, their corresponding sides are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q4/5

494

QUESTION ID

S15-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q4/5

True or false?

Any two right triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q5/5

495

QUESTION ID

S15-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 8.EE.B.6

FORWARD 2

Use slopes of parallel and perpendicular lines • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Line p has a slope of $\frac{3}{4}$.

What is the slope of a line perpendicular to p?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q1/5

497

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−4/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q1/5

Write an equation of the line through (0, 2) that is parallel to y = −3x + 7.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q2/5

498

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = −3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q2/5

Which line is perpendicular to $y = 2x - 5$?

A. $y = -2x - 5$

B. $y = 2x + 1$

C. $y = -\frac{1}{2}x + 1$

D. $y = \frac{1}{2}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q3/5

499

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = -(1/2)x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

Parallel lines that are not vertical have equal slopes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q4/5

500

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q4/5

True or false?

Lines with slopes 3 and −3 are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q5/5

501

QUESTION ID

S15-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular slopes multiply to −1; 3 × (−3) = −9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q5**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q5/5

SET 60 • 8.EE.B.6 • EXPRESSIONS & EQUATIONS

Use similar triangles to derive y = mx for a line through the origin

MAIN	Grade 8 main question	**8.EE.B.6**
BACKWARD 1	Represent a proportional relationship with an equation	**7.RP.A.2.c**
BACKWARD 2	Find the constant of proportionality from a point or a table	**7.RP.A.2.b**
FORWARD 1	Show that a dilation centered at the origin maps a line through th…	**HSG-SRT.A.1**
FORWARD 2	Use rotated slope triangles to explain the slopes of parallel and p…	**HSG-GPE.B.5**

SET 60 • 8.EE.B.6

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.6

A line passes through the origin and (4, 6). For any other point (x, y) on the line (x ≠ 0), the triangle with vertices (0, 0), (x, 0), (x, y) is similar to the triangle with vertices (0, 0), (4, 0), (4, 6).

Use this to write the equation of the line.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q1/5

504

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

y = 1.5x (because y ÷ x = 6 ÷ 4 for x ≠ 0, and the origin also fits)

GRADING NOTE y = (3/2)x is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q1/5

A line passes through the origin with slope m. Use a slope triangle from (0, 0) to a point (x, y) on the line ($x \neq 0$) to explain why every point on the line satisfies $y = mx$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q2/5

505

QUESTION ID

S60-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

y = mx

GRADING NOTE REQUIRED WORK: The slope triangle with vertices (0, 0), (x, 0), (x, y) is similar to the triangle with run 1 and rise m, so $y \div x = m \div 1$ and $y = mx$. The point (0, 0) also satisfies $y = m \cdot 0$. Any equivalent explanation is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q2/5

A line through the origin contains the point (5, 2).

Which equation does a similar-triangle argument give for the line?

A. $y = 2x + 5$

B. $y = x + 2$

C. $y = \frac{5}{2}x$

D. $y = \frac{2}{5}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q3/5

506

QUESTION ID

S60-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = (2/5)x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q3/5

A line through the origin has a slope of −3. The point (2, y) is on the line.

Use a slope triangle to find y, then write the equation of the line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q4/5

507

QUESTION ID

S60-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = −6; the line is y = −3x

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q4/5

True or false?

The point (6, 4) is on the line through the origin and (3, 2), because $4 \div 6 = 2 \div 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q5/5

508

QUESTION ID

S60-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q5/5

SET 60 • 8.EE.B.6

BACKWARD 1

Represent a proportional relationship with an equation • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Each notebook costs $3.

Write an equation for the cost c of n notebooks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q1/5

510

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 3n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	2	5	8
y	5	12.5	20

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q2/5

511

QUESTION ID

S60-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2.5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q2/5

A car goes 28 miles on each gallon of gas.

Which equation gives the miles m for g gallons?

A. m = 28g

B. g = 28m

C. m = 28 ÷ g

D. m = g + 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q3/5

512

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. m = 28g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q3/5

True or false?

The equation y = 7x describes a proportional relationship with a constant of proportionality of 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q4/5

513

QUESTION ID

S60-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q4/5

True or false?

The equation y = x + 4 describes a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q5/5

514

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its graph does not pass through the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q5/5

SET 60 • 8.EE.B.6

BACKWARD 2

Find the constant of proportionality from a point or a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A proportional relationship passes through (6, 9).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q1/5

516

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 3/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a proportional relationship.

What is the constant of proportionality?

x	4	10
y	2	5

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q2/5

517

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.5

GRADING NOTE 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q2/5

What is the constant of proportionality in y = 12x?

A. 6

B. $\frac{1}{12}$

C. 12

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q3/5

518

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q3/5

True or false?

The graph of a proportional relationship with constant k passes through (1, k).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q4/5

519

QUESTION ID

S60-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q4/5

True or false?

A line through (0, 0) and (4, 10) has a constant of proportionality of 0.4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q5/5

520

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $k = 10 \div 4 = 2.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q5/5

SET 60 • 8.EE.B.6

FORWARD 1

Show that a dilation centered at the origin maps a line through the origin onto itself • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A dilation with center (0, 0) and a scale factor of 3 is applied to the point (2, 5) on the line y = 2.5x.

What is the image, and is it on the same line? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q1/5

522

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(6, 15); yes, because 15 = 2.5 · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q1/5

Explain why a dilation centered at the origin with scale factor k (k > 0) maps the line y = mx onto itself.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q2/5

523

QUESTION ID

S60-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

The image of the line y = mx is the line y = mx.

GRADING NOTE REQUIRED WORK: A point (x, mx) maps to (kx, kmx) = (X, mX) with X = kx, which is on y = mx; every point of the line is the image of a point of the line. Any equivalent explanation is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q2/5

A dilation centered at (0, 0) with a scale factor of 2 is applied to the line y = 3x + 1.

What is the image?

A. y = 3x + 1

B. y = 3x + 2

C. y = 6x + 2

D. y = 6x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q3/5

524

QUESTION ID

S60-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q3/5

True or false?

A dilation centered at the origin maps every line through the origin onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q4/5

525

QUESTION ID

S60-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q4/5

True or false?

A dilation always maps a line onto a line with a different slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q5/5

526

QUESTION ID

S60-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A dilation maps a line to a parallel line or to the same line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q5/5

SET 60 • 8.EE.B.6

FORWARD 2

Use rotated slope triangles to explain the slopes of parallel and perpendicular lines • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

A slope triangle for line p has a run of 3 and a rise of 2. Rotating it 90° counterclockwise gives a slope triangle for a line q perpendicular to p.

What are the run and rise of the rotated triangle, and what is the slope of q?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q1/5

528

QUESTION ID

S60-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Run −2 and rise 3 (or run 2 and rise −3); the slope of q is −3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q1/5

Use slope triangles to explain why two parallel, non-vertical lines have equal slopes.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q2/5

529

QUESTION ID

S60-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A transversal makes equal corresponding angles with the two lines, so their slope triangles have the same angles and are similar; similar triangles have equal rise ÷ run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q2/5

Line p has a slope of $\frac{4}{5}$.

What is the slope of a line perpendicular to p?

A. $-\frac{5}{4}$

B. $\frac{5}{4}$

C. $-\frac{4}{5}$

D. $\frac{4}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q3/5

530

QUESTION ID

S60-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −5/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q3/5

True or false?

Lines with slopes 2 and $-\frac{1}{2}$ are perpendicular, because $2 \times (-\frac{1}{2}) = -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q4/5

531

QUESTION ID

S60-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q4/5

True or false?

Two lines whose slopes multiply to 1 are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q5/5

532

QUESTION ID

S60-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q5/5

SET 61 • 8.EE.B.6 • EXPRESSIONS & EQUATIONS

Use similar triangles to derive y = mx + b for a line that crosses the y-axis at b

MAIN	Grade 8 main question	**8.EE.B.6**
BACKWARD 1	Find horizontal and vertical distances between points	**6.NS.C.8**
BACKWARD 2	Solve equations of the form px + q = r	**7.EE.B.4.a**
FORWARD 1	Construct linear functions from a point and a slope or from two po…	**HSF-LE.A.2**
FORWARD 2	Describe vertical and horizontal translations of linear functions	**HSF-BF.B.3**

SET 61 • 8.EE.B.6

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.6

A line has a slope of 2 and crosses the y-axis at (0, 3).

Use a slope triangle from (0, 3) to a point (x, y) on the line ($x \neq 0$) to explain why $y = 2x + 3$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q1/5

535

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

The slope triangle has run x and rise y − 3, so (y − 3) ÷ x = 2. Then y − 3 = 2x and y = 2x + 3. The point (0, 3) also fits.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q1/5

The line y = mx is translated b units up.

Explain why the image is the line y = mx + b.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q2/5

536

QUESTION ID

S61-M-Q2

TYPE

Short answer

CORRECT ANSWER

Each point (x, mx) moves to (x, mx + b), so every point of the image satisfies y = mx + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q2/5

A line crosses the y-axis at (0, −4) and has a slope of $\frac{1}{2}$.

For a point (x, y) on the line with x ≠ 0, which equation does the slope triangle from (0, −4) give?

A. $(y - 4) \div x = \frac{1}{2}$

B. $(y + 4) \div x = 2$

C. $(y + 4) \div x = \frac{1}{2}$

D. $y \div x = \frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q3/5

537

QUESTION ID

S61-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (y + 4) ÷ x = 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q3/5

A line crosses the y-axis at (0, −2) and passes through (5, 8).

Use a slope triangle to find the slope, then write the equation in the form y = mx + b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q4/5

538

QUESTION ID

S61-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = 2x − 2 (the slope triangle has run 5 and rise 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q4/5

True or false?

The line y = 3x + 4 is the image of y = 3x after a translation 4 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q5/5

539

QUESTION ID

S61-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the image after a translation 4 units up; 4 units right gives y = 3x − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q5/5

SET 61 • 8.EE.B.6

BACKWARD 1

Find horizontal and vertical distances between points • Grade 6 • The Number System • 6.NS.C.8

How far is it horizontally from (0, 3) to (4, 3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q1/5

541

QUESTION ID

S61-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q1/5

How far is it vertically from (4, 3) to (4, 11)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q2/5

542

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q2/5

What is the distance between (2, −1) and (2, 5)?

A. 3 units

B. 7 units

C. 4 units

D. 6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q3/5

543

QUESTION ID

S61-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q3/5

True or false?

The distance between (−3, 2) and (4, 2) is 7 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q4/5

544

QUESTION ID

S61-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q4/5

True or false?

The distance between (1, −2) and (1, 6) is 4 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q5/5

545

QUESTION ID

S61-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 8 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • 8.EE.B.6

BACKWARD 2

Solve equations of the form px + q = r • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for b.

$11 = 3(4) + b$

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q1/5

547

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q1/5

Solve for x.

$7 = 2x + 3$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q2/5

548

QUESTION ID

S61-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q2/5

Solve for b.

$5 = -2(3) + b$

A. b = 11

B. b = −1

C. b = 1

D. b = −11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q3/5

549

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. b = 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q3/5

True or false?

If 9 = 4(2) + b, then b = 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q4/5

550

QUESTION ID

S61-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q4/5

True or false?

If 2 = 5(0) + b, then b = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q5/5

551

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE b = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q5/5

SET 61 • 8.EE.B.6

FORWARD 1

Construct linear functions from a point and a slope or from two points • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the equation of the line through (2, 5) with a slope of 3 in point-slope form, then in slope-intercept form.

Equations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q1/5

553

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y − 5 = 3(x − 2);
y = 3x − 1

GRADING NOTE Both forms are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q1/5

Explain why every point (x, y) on the line through (2, 5) with a slope of 3 satisfies (y − 5) ÷ (x − 2) = 3 when x ≠ 2.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q2/5

554

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The slope triangle from (2, 5) to (x, y) has run x − 2 and rise y − 5; every slope triangle on the line gives the same ratio, 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q2/5

Which equation describes the line through (2, 1) and (5, 10)?

A. $y = 9x - 17$

B. $y = 3x + 1$

C. $y = 3x - 5$

D. $y = \frac{1}{3}x + \frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q3/5

555

QUESTION ID

S61-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 3x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q3/5

True or false?

The line $y - 2 = -4(x + 1)$ passes through $(-1, 2)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q4/5

556

QUESTION ID

S61-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q4/5

True or false?

The line through (0, 0) and (2, 6) is y = 2x + 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S61-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q5/5

557

QUESTION ID

S61-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is y = 3x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S61-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q5/5

SET 61 • 8.EE.B.6

FORWARD 2

Describe vertical and horizontal translations of linear functions • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3

$f(x) = 2x$ and $g(x) = f(x) + 4$.

Write g(x), and describe how the graph of g compares with the graph of f.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q1/5

559

QUESTION ID

S61-F2-Q1

TYPE

Short answer

CORRECT ANSWER

g(x) = 2x + 4; the graph of f is translated 4 units up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q1/5

$f(x) = 2x$ and $h(x) = f(x - 3)$.

Write h(x) in slope-intercept form, and describe the translation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q2/5

560

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

h(x) = 2x − 6; the graph of f is translated 3 units right (which, for this line, is the same as 6 units down).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q2/5

f(x) = −x + 1. Which function translates the graph of f 5 units down?

A. g(x) = −x + 1 − 5x

B. g(x) = −(x − 5) + 1

C. g(x) = −x + 6

D. g(x) = −x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q3/5

561

QUESTION ID

S61-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. g(x) = −x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q3/5

True or false?

For a line f(x) = mx + b, f(x − 2) = f(x) − 2m for every x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q4/5

562

QUESTION ID

S61-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE m(x − 2) + b = mx + b − 2m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q4/5

True or false?

g(x) = f(x) − 5 moves the graph of f 5 units to the left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q5/5

563

QUESTION ID

S61-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It moves the graph 5 units down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q5/5

SET 16 • 8.EE.C.7.a • EXPRESSIONS & EQUATIONS

Decide whether a linear equation has one solution, no solution, or infinitely many

MAIN	Grade 8 main question	**8.EE.C.7.a**
BACKWARD 1	Expand linear expressions	**7.EE.A.1**
BACKWARD 2	Solve two-step equations	**7.EE.B.4.a**
FORWARD 1	Find parameter values that give a system one solution, no solution,	**HSA-REI.C.6**
FORWARD 2	Use the form $(x - p)^2 = q$ to find how many real solutions a quadra	**HSA-REI.B.4.a**

SET 16 • 8.EE.C.7.a

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$4(x + 3) = 4x + 12$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q1/5

566

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q1/5

How many solutions does this equation have?

$5x - 2 = 5x + 7$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q2/5

567

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q2/5

Which equation has exactly one solution?

A. $3(x + 1) = 3x + 3$

B. $2x - 2x = 0$

C. $3x + 1 = 2x + 9$

D. $3x + 1 = 3x + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q3/5

568

QUESTION ID

S16-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3x + 1 = 2x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q3/5

What number in the blank makes the equation have no solution?

$6x + 5 = ___x + 1$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q4/5

569

QUESTION ID

S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q4/5

True or false?

$2(x - 4) = 2x - 4$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q5/5

570

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It simplifies to $-8 = -4$, so it has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q5/5

SET 16 • 8.EE.C.7.a

BACKWARD 1

Expand linear expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$4(x + 3)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q1/5

572

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4x + 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q1/5

Expand.

$-2(3x - 5)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q2/5

573

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-6x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q2/5

Which expression is equal to 5(2x − 1)?

A. 10x + 5

B. 10x − 1

C. 10x − 5

D. 7x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q3/5

574

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$3(x + 2) = 3x + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q4/5

575

QUESTION ID

S16-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$-(x - 4) = -x - 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q5/5

576

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 8.EE.C.7.a

BACKWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

3x + 7 = 22

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q1/5

578

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q1/5

Solve for x.

$-2x + 4 = 10$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q2/5

579

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q2/5

Solve for x.

$5x - 6 = 19$

A. $x = \frac{13}{5}$

B. $x = -5$

C. $x = 5$

D. $x = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q3/5

580

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x = 2$ is the solution of $4x + 1 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q4/5

581

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q4/5

True or false?

$x = 3$ is the solution of $2x - 5 = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q5/5

582

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 8.EE.C.7.a

FORWARD 1

Find parameter values that give a system one solution, no solution, or infinitely many • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

For which value of k does the system have no solution?

$y = kx + 2$

$y = 3x - 5$

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q1/5

584

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q1/5

For which values of a and b does the system have infinitely many solutions?

$2x + 3y = 6$

$ax + 6y = b$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q2/5

585

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 4 and b = 12

GRADING NOTE Both values are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q2/5

For which value of c does the system have infinitely many solutions?

$4x - 2y = c$

$y = 2x - 3$

A. -6 **B.** 6

C. -3 **D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q3/5

586

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q3/5

True or false?

The system y = mx + 1 and y = 2x + 1 has exactly one solution for every value of m except 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q4/5

587

QUESTION ID

S16-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q4/5

True or false?

If two lines have different y-intercepts, the system they form has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q5/5

588

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Lines with different slopes cross once, whatever their y-intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 8.EE.C.7.a

FORWARD 2

Use the form $(x - p)^2 = q$ to find how many real solutions a quadratic equation has • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

How many real solutions does $(x - 3)^2 = 0$ have? Give any solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q1/5

590

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

One: x = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q1/5

How many real solutions does $(x + 1)^2 = -4$ have? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q2/5

591

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

None. A square of a real number cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q2/5

Completing the square on $x^2 + 6x + 13 = 0$ gives $(x + 3)^2 = q$.

What is q, and how many real solutions are there?

A. q = 4; two real solutions

B. q = −4; one real solution

C. q = 13; two real solutions

D. q = −4; no real solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q3/5

592

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. q = −4; no real solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q3/5

True or false?

$(x - 2)^2 = 9$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q4/5

593

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x = 5$ and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q4/5

True or false?

$x^2 - 6x + 9 = 0$ has two different real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q5/5

594

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is $(x - 3)^2 = 0$, so $x = 3$ is the only solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q5/5

SET 62 • 8.EE.C.7.a • EXPRESSIONS & EQUATIONS

Give examples of linear equations with one, no, or infinitely many solutions and show which by transforming them

MAIN	Grade 8 main question	**8.EE.C.7.a**
BACKWARD 1	Simplify expressions with the distributive property	**7.EE.A.1**
BACKWARD 2	Decide whether a value makes an equation true	**6.EE.B.5**
FORWARD 1	Find the parameter values that give a linear equation one, no, or i…	**HSA-REI.B.3**
FORWARD 2	Justify each step and state the solution set of an equation	**HSA-REI.A.1**

SET 62 • 8.EE.C.7.a

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Write a linear equation in x, with x on both sides, that has no solution. Transform it to show why.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q1/5

597

QUESTION ID

S62-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Any correct example, such as 2x + 3 = 2x + 5 (3 = 5 is false)

GRADING NOTE REQUIRED WORK: Any equation that transforms to a false statement a = b with a ≠ b. Example: 2x + 3 = 2x + 5 becomes 3 = 5 after subtracting 2x. The steps must be shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q1/5

Write a linear equation in x, with x on both sides, that has infinitely many solutions. Transform it to show why.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q2/5

598

QUESTION ID

S62-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Any correct example, such as 3(x + 1) = 3x + 3 (3 = 3 is always true)

GRADING NOTE REQUIRED WORK: Any equation that transforms to a true statement a = a. Example: 3(x + 1) = 3x + 3 becomes 3x + 3 = 3x + 3, then 3 = 3. The steps must be shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q2/5

Write a linear equation in x, with x on both sides, whose only solution is x = 4. Transform it to x = 4.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q3/5

599

QUESTION ID

S62-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Any correct example, such as 5x − 2 = 3x + 6

GRADING NOTE REQUIRED WORK: Any equation with different coefficients of x that transforms to x = 4. Example: 5x − 2 = 3x + 6; 2x = 8; x = 4. The steps must be shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q3/5

Fill in the blanks so that the equation has infinitely many solutions.

$4(x - 2) = ___x - ___$

Blanks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q4/5

600

QUESTION ID

S62-M-Q4

TYPE

Short answer

CORRECT ANSWER

4 and 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q4/5

Ana transformed an equation and got 7 = 7.

Which equation could she have started with?

A. 2x + 7 = 3x + 7

B. x + 7 = 0

C. 2x + 3 = 2x + 7

D. 2(x + 3) + 1 = 2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q5/5

601

QUESTION ID

S62-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D.

2(x + 3) + 1 = 2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q5/5

SET 62 • 8.EE.C.7.a

BACKWARD 1

Simplify expressions with the distributive property • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$2(x + 3) + 1$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q1/5

603

QUESTION ID

S62-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q1/5

Simplify.

$5x - 2 - 3x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q2/5

604

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2x − 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q2/5

Which expression is equal to 4(x − 2)?

A. 4x − 2

B. 4x − 8

C. x − 8

D. 4x + 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q3/5

605

QUESTION ID

S62-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4x − 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q3/5

True or false?

$3x + 2x - 1 = 5x - 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q4/5

606

QUESTION ID

S62-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q4/5

True or false?

$-(2x - 3) = -2x - 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q5/5

607

QUESTION ID

S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-(2x - 3) = -2x + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • 8.EE.C.7.a

BACKWARD 2

Decide whether a value makes an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 3 a solution of 2x + 1 = x + 4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q1/5

609

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Both sides equal 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q1/5

Which of the values 0, 1, and 2 makes $3x - 1 = x + 3$ true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q2/5

610

QUESTION ID

S62-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q2/5

Which value makes 4x = 2x + 6 true?

A. 3

B. 4

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q3/5

611

QUESTION ID

S62-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q3/5

True or false?

Every value of x makes x + x = 2x true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q4/5

612

QUESTION ID

S62-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q4/5

True or false?

$x = 5$ makes $x + 2 = 2x - 2$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q5/5

613

QUESTION ID

S62-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $7 \neq 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • 8.EE.C.7.a

FORWARD 1

Find the parameter values that give a linear equation one, no, or infinitely many solutions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

For which value of a does the equation ax + 3 = 5x + 1 have no solution?

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q1/5

615

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q1/5

For which values of a and b does ax + b = 3x − 2 have infinitely many solutions?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q2/5

616

QUESTION ID

S62-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 3 and b = −2

GRADING NOTE Both values are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q2/5

For which values of c does $cx - 4 = 2x + 4$ have exactly one solution?

A. Every value of c

B. No value of c

C. Every value except c = 2

D. Only c = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q3/5

617

QUESTION ID

S62-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Every value except c = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q3/5

True or false?

When $a \neq 4$, the equation $ax + 1 = 4x + 9$ has the one solution $x = 8 \div (a - 4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q4/5

618

QUESTION ID

S62-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q4/5

True or false?

When k = 0, the equation kx = kx + 1 has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q5/5

619

QUESTION ID

S62-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It becomes 0 = 1, which has no solution (for every k).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q5/5

SET 62 • 8.EE.C.7.a

FORWARD 2

Justify each step and state the solution set of an equation • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Transform 2(x + 4) − 3 = 2x + 5, giving a reason for each step, and state the solution set.

Solution set: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q1/5

621

QUESTION ID

S62-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

All real numbers

GRADING NOTE REQUIRED WORK: Every step with a valid reason. Other valid orders of steps are accepted. A correct answer without reasons is not full credit. Example: distributive property (2x + 8 − 3 = 2x + 5); combine like terms (2x + 5 = 2x + 5); subtract 2x from both sides (5 = 5), which is always true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q1/5

Transform 6x − 1 = 2(3x + 2), giving a reason for each step, and explain why the equation has no solution.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q2/5

622

QUESTION ID

S62-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

No solution

GRADING NOTE REQUIRED WORK: Every step with a valid reason. Other valid orders of steps are accepted. A correct answer without reasons is not full credit. Example: distributive property (6x − 1 = 6x + 4); subtract 6x from both sides (−1 = 4), which is never true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q2/5

An equation transforms to 0 = 0.

What is its solution set?

A. All real numbers

B. Only x = 1

C. No solution

D. Only x = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q3/5

623

QUESTION ID

S62-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. All real numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q3/5

True or false?

Adding the same expression, such as −3x, to both sides of an equation keeps the same solution set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q4/5

624

QUESTION ID

S62-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q4/5

True or false?

Dividing both sides of $0 \cdot x = 0$ by 0 shows that its only solution is $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q5/5

625

QUESTION ID

S62-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined; every value of x is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q5/5

SET 17 • 8.EE.C.7.b • EXPRESSIONS & EQUATIONS

Solve multistep linear equations by expanding and collecting like terms

MAIN	Grade 8 main question	**8.EE.C.7.b**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Solve two-step equations with rational numbers	**7.EE.B.4.a**
FORWARD 1	Solve linear equations with letter coefficients	**HSA-REI.B.3**
FORWARD 2	Justify each step of a solution and find an invalid step	**HSA-REI.A.1**

SET 17 • 8.EE.C.7.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$3(2x - 4) = 4x + 6$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q1/5

628

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q1/5

Solve for x.

$5 - 2(x + 1) = 3x - 7$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q2/5

629

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q2/5

Solve for x.

$4(x - 1) - (x + 2) = 2x + 3$

A. $x = 1$

B. $x = 5$

C. $x = 9$

D. $x = -9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q3/5

630

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve. Show each step.

$-3(x + 5) + 2x = 4(x - 10)$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q4/5

631

QUESTION ID

S17-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

5

GRADING NOTE REQUIRED WORK: Any valid sequence of steps. Example: $-3x - 15 + 2x = 4x - 40$; $-x - 15 = 4x - 40$; $25 = 5x$; $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q4/5

True or false?

$x = -1$ is the solution of $2(3x + 1) = 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q5/5

632

QUESTION ID

S17-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q5/5

SET 17 • 8.EE.C.7.b

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$6x - 4 + 2x + 9$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q1/5

634

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q1/5

Simplify.

$-3a + 7 - a$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q2/5

635

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-4a + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q2/5

Simplify.

$2(x + 3) + 5x$

A. $10x + 6$

B. $2x + 11$

C. $7x + 3$

D. $7x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q3/5

636

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q3/5

True or false?

$4y - y = 3y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q4/5

637

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q4/5

True or false?

$5n + 2 - 2n = 3n$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q5/5

638

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5n + 2 - 2n = 3n + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 8.EE.C.7.b

BACKWARD 2

Solve two-step equations with rational numbers • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$6x - 5 = 31$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q1/5

640

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q1/5

Solve for x.

$\frac{x}{4} + 3 = 7$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q2/5

641

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q2/5

Solve for x.

$-3x + 8 = -10$

A. $x = -6$

B. $x = -\frac{2}{3}$

C. $x = \frac{2}{3}$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q3/5

642

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x = -2$ is the solution of $5x + 12 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q4/5

643

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q4/5

True or false?

$x = 4$ is the solution of $3x - 2 = 14$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q5/5

644

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 8.EE.C.7.b

FORWARD 1

Solve linear equations with letter coefficients • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x. Assume a ≠ 3.

ax + 2 = 3x + 8

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q1/5

646

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6/(a − 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q1/5

Solve for x. Assume k ≠ 0.

$k(x - 4) = 2k$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q2/5

647

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q2/5

Solve for x. Assume b ≠ 0.

$bx - 5b = 2b$

A. $x = 7b$

B. $x = \frac{7}{b}$

C. $x = 7$

D. $x = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q3/5

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q3/5

True or false?

If $c \neq 1$, the solution of $cx + 1 = x + 1$ is $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q4/5

649

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q4/5

True or false?

The solution of $mx = m + 4$ is $x = 5$ for every nonzero value of m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q5/5

650

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = (m + 4)/m$, which is 5 only when $m = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 8.EE.C.7.b

FORWARD 2

Justify each step of a solution and find an invalid step • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2(x − 3) + 4 = 5x − 11. Give a reason for each step.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q1/5

652

QUESTION ID

S17-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

3

GRADING NOTE REQUIRED WORK: Every step with a valid reason. Other valid orders of steps are accepted. A correct answer without reasons is not full credit. Example: distributive property (2x − 6 + 4 = 5x − 11); combine like terms (2x − 2 = 5x − 11); subtract 2x and add 11 to both sides (9 = 3x); divide both sides by 3 (x = 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q1/5

Dee solved $4 - 2(x - 1) = 8$ like this:

Step 1: $4 - 2x - 2 = 8$

Step 2: $2 - 2x = 8$

Step 3: $x = -3$

Which step is wrong? Correct it and give the solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q2/5

653

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Step 1: $-2(x - 1) = -2x + 2$, so $6 - 2x = 8$ and $x = -1$.

GRADING NOTE Grade both: Step 1 with the correction, and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q2/5

Which reason justifies rewriting $3x - 5 = 7$ as $3x = 12$?

A. Addition property of equality

B. Multiplication property of equality

C. Commutative property of addition

D. Distributive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q3/5

654

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Addition property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q3**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q3/5

True or false?

Each step of a valid solution gives an equation with the same solutions as the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q4/5

655

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q4/5

True or false?

Multiplying both sides of an equation by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q5/5

656

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q5/5

SET 18 • 8.EE.C.7.b • EXPRESSIONS & EQUATIONS

Write and solve linear equations with rational coefficients for real-world problems

MAIN	Grade 8 main question	**8.EE.C.7.b**
BACKWARD 1	Solve word problems that lead to px + q = r	**7.EE.B.4.a**
BACKWARD 2	Multiply rational numbers in context	**7.NS.A.3**
FORWARD 1	Solve real-world systems, including mixtures and rates, and judge…	**HSA-REI.C.6**
FORWARD 2	Create equations and inequalities in one variable to solve proble…	**HSA-CED.A.1**

SET 18 • 8.EE.C.7.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Gym A charges $30 to join plus $12.50 per month. Gym B charges $55 to join plus $10 per month.

Write and solve an equation to find when the total costs are equal.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q1/5

659

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

The equation 30 + 12.5m = 55 + 10m and the answer m = 10 months

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Plant A is $6\frac{1}{2}$ inches tall and grows $\frac{3}{4}$ inch per week.

Plant B is 10 inches tall and grows $\frac{1}{4}$ inch per week.

Write and solve an equation to find when they are the same height.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q2/5

660

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 6.5 + 0.75w = 10 + 0.25w and the answer w = 7 weeks

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q2/5

Solve for x.

$\frac{2}{3}x - 4 = \frac{1}{2}x + 1$

A. x = 18

B. $x = \frac{30}{7}$

C. x = 30

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q3/5

661

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q3/5

Solve for x.

$0.4(x + 15) = 0.25x + 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q4/5

662

QUESTION ID

S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q4/5

True or false?

$x = 12$ is the solution of $\frac{3}{4}x + 2 = \frac{1}{2}x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q5/5

663

QUESTION ID

S18-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q5/5

SET 18 • 8.EE.C.7.b

BACKWARD 1

Solve word problems that lead to px + q = r • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Ana has $40. She buys some $6.50 tickets and has $1 left.

How many tickets did she buy?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q1/5

665

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q1/5

A rental costs $20 plus $8 per hour. The total is $68.

For how many hours was it rented?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q2/5

666

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q2/5

Solve for x.

0.5x + 3 = 11

A. x = 7

B. x = 16

C. x = 28

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q3/5

667

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q3/5

True or false?

$x = 9$ is the solution of $\frac{1}{3}x + 2 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q4/5

668

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q4/5

True or false?

$x = 4$ is the solution of $2.5x - 1 = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q5/5

669

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 4.8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 8.EE.C.7.b

BACKWARD 2

Multiply rational numbers in context • Grade 7 • The Number System • 7.NS.A.3

A plant grows $\frac{3}{4}$ inch per week.

How much does it grow in 8 weeks?

Growth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q1/5

671

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q1/5

A gym charges $12.50 per month.

How much do 10 months cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q2/5

672

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q2/5

A cyclist rides 0.4 mile per minute for 15 minutes.

How far does she ride?

A. 6 miles

B. 60 miles

C. 4 miles

D. 15.4 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q3/5

673

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3}$ of 9 feet is 6 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q4/5

674

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

0.25 of a 12-hour day is 4 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q5/5

675

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.25 × 12 = 3 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 8.EE.C.7.b

FORWARD 1

Solve real-world systems, including mixtures and rates, and judge whether a solution makes sense • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

A chemist mixes a 10% salt solution with a 40% salt solution to make 60 liters of a 20% salt solution.

Write and solve a system to find how many liters of each she uses.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q1/5

677

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The system x + y = 60 and 0.10x + 0.40y = 12, and the answer 40 liters of the 10% solution and 20 liters of the 40% solution

GRADING NOTE Both parts are required: a valid system and the correct solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q1/5

A ticket system gives a = 37.5 adult tickets and s = 82.5 student tickets.

Does this solution make sense? Explain what it tells you.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q2/5

678

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. Ticket counts must be whole numbers, so the given totals (or the model) cannot be right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q2/5

Plan A costs $y = 12x + 40$ dollars and Plan B costs $y = 20x$ dollars for x months.

For which numbers of months is Plan A cheaper?

A. Exactly 5 months

B. Fewer than 5 months

C. More than 5 months

D. Never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q3/5

679

QUESTION ID

S18-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. More than 5 months

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q3/5

True or false?

A plane flies 600 miles in 2 hours with the wind and 600 miles in 3 hours against the wind. Its speed in still air is 250 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q4/5

680

QUESTION ID

S18-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q4/5

True or false?

A system for numbers of bikes b and cars c with the solution b = −4, c = 20 gives a sensible answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q5/5

681

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A count of bikes cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q5/5

SET 18 • 8.EE.C.7.b

FORWARD 2

Create equations and inequalities in one variable to solve problems • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1

Plan A costs $25 plus $0.15 per text. Plan B costs $40 with unlimited texts.

Write and solve an inequality to find when Plan A costs less.

Inequality and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q1/5

683

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The inequality 25 + 0.15t < 40 and the answer t < 100 texts (0 to 99 texts)

GRADING NOTE Both parts are required: a valid inequality and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q1/5

Two trains leave a station at the same time in opposite directions, at 55 and 65 miles per hour.

Write and solve an equation to find when they are 300 miles apart.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q2/5

684

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 55t + 65t = 300 and the answer t = 2.5 hours

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q2/5

A rectangle's length is 3 more than twice its width w. Its perimeter is 54.

Which equation can find w?

A. 2(2w) + 3 = 54

B. w + 2w + 3 = 54

C. 2w + 2(2w + 3) = 54

D. w(2w + 3) = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q3/5

685

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C.

2w + 2(2w + 3) = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q3/5

True or false?

A rectangle has a length of 2w + 3 and a perimeter of 54. Its width w is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q4/5

686

QUESTION ID

S18-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q4/5

True or false?

The inequality $25 + 0.15t < 40$ is true when $t = 120$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q5/5

687

QUESTION ID
S18-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $25 + 18 = 43$, which is not less than 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q5/5

SET 19 • 8.EE.C.8.a • EXPRESSIONS & EQUATIONS

Solve a system of two linear equations by graphing

MAIN	Grade 8 main question	**8.EE.C.8.a**
BACKWARD 1	Make a table and graph an equation	**6.EE.C.9**
BACKWARD 2	Graph points in all four quadrants	**6.NS.C.8**
FORWARD 1	Solve f(x) = g(x) from the intersection of two graphs	**HSA-REI.D.11**
FORWARD 2	Graph the solution set of a system of linear inequalities	**HSA-REI.D.12**

SET 19 • 8.EE.C.8.a

MAIN

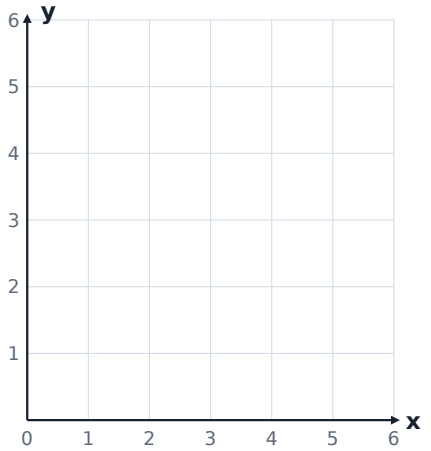
Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph y = x + 1 and
y = −x + 5 on the grid.

What is the solution of the system?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q1/5

690

QUESTION ID
S19-M-Q1

TYPE
Drawing + written answer

CORRECT ANSWER
Drawing: The line y = x + 1 through (0, 1) and (4, 5), and the line y = −x + 5 through (0, 5) and (5, 0), crossing at (2, 3)

Written: (2, 3)

GRADING NOTE Grade both: the two lines drawn correctly, and the solution (2, 3).

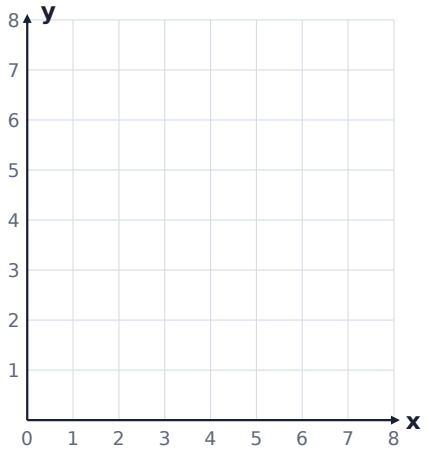
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q1/5

Graph y = 0.5x + 1 and y = −x + 7 on the grid.

What is the solution of the system?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q2/5

691

QUESTION ID

S19-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The line y = 0.5x + 1 through (0, 1) and (6, 4), and the line y = −x + 7 through (0, 7) and (7, 0), crossing at (4, 3)

Written: (4, 3)

GRADING NOTE Grade both: the two lines drawn correctly, and the solution (4, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

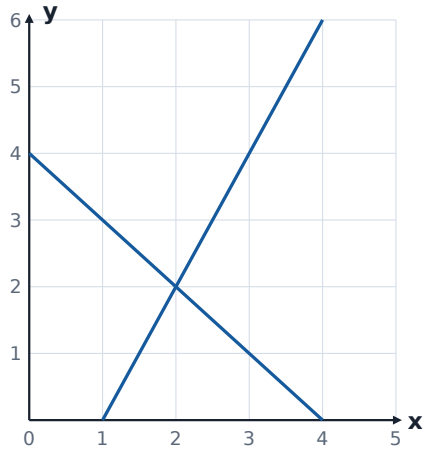
Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows the lines $y = -x + 4$ and $y = 2x - 2$.

What is the solution of the system? Check it in both equations.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q3/5

692

QUESTION ID

S19-M-Q3

TYPE

Short answer

CORRECT ANSWER

(2, 2): 2 = −2 + 4 and 2 = 2(2) − 2.

GRADING NOTE Grade both: (2, 2) and the check in both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q3/5

The graphs of the two equations in a system are parallel lines.

What does this tell you about the system?

A. It has no solution.

B. It has infinitely many solutions.

C. It has one solution.

D. Its solution is the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q4/5

693

QUESTION ID

S19-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q4/5

True or false?

When the graphs of two linear equations are two different lines that cross, the crossing point is the only solution of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q5/5

694

QUESTION ID

S19-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q5/5

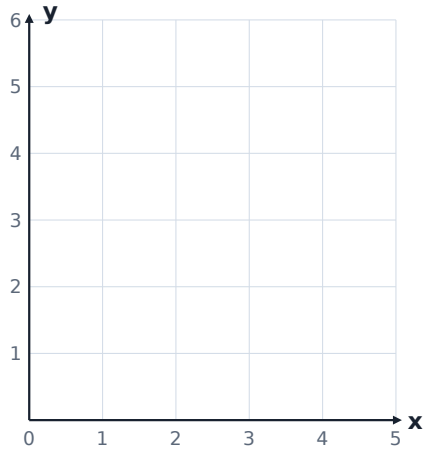
SET 19 • 8.EE.C.8.a

BACKWARD 1

Make a table and graph an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

Make a table for y = x + 2 with x = 0, 1, 2, and 3. Write the table, then graph the points on the grid.

Table: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q1/5

696

QUESTION ID

S19-B1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points at (0, 2), (1, 3), (2, 4), and (3, 5)

Written: x: 0, 1, 2, 3; y: 2, 3, 4, 5

GRADING NOTE Grade both: the table of all four pairs, and the four points graphed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q1/5

What is the value of y when x = 4 in y = 3x?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q2/5

697

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q2/5

Which point is on the graph of y = x + 3?

A. (1, 3)

B. (3, 0)

C. (5, 2)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q3/5

698

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q3/5

True or false?

(4, 8) is on the graph of y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q4/5

699

QUESTION ID

S19-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q4/5

True or false?

(0, 1) is on the graph of y = x + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q5/5

700

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

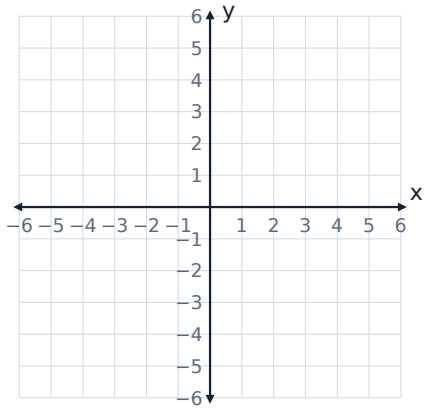
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 8.EE.C.8.a

BACKWARD 2

Graph points in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Plot A(2, −3) and B(−4, 1) on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q1/5

QUESTION ID

S19-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Point A at (2, −3) and point B at (−4, 1)

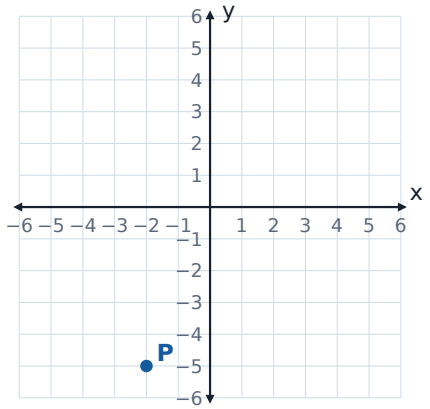
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q1/5

What are the coordinates of point P?

Coordinates: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q2/5

703

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q2/5

Which point is in Quadrant II?

A. (3, −4)

B. (−3, −4)

C. (3, 4)

D. (−3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q3/5

704

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

(0, −2) is on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q4/5

705

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

(5, −1) is in Quadrant III.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q5/5

706

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (5, −1) is in Quadrant IV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 8.EE.C.8.a

FORWARD 1

Solve f(x) = g(x) from the intersection of two graphs • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of f(x) = x + 1 and g(x) = 7 − x intersect at (3, 4).

What is the solution of x + 1 = 7 − x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q1/5

708

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q1/5

Use a table of values for x = 0, 1, 2, and 3 to solve 2x + 1 = 7 − x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q2/5

709

QUESTION ID

S19-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

2

GRADING NOTE REQUIRED WORK: A table with both sides for each x: 2x + 1 is 1, 3, 5, 7 and 7 − x is 7, 6, 5, 4. The sides are equal at x = 2 (both 5). The table must be shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q2/5

The graphs of $f(x) = x^2$ and $g(x) = x + 2$ intersect at (2, 4) and (−1, 1).

What are the solutions of $x^2 = x + 2$?

A. x = 2 only

B. x = −1 only

C. x = 4 and x = 1

D. x = 2 and x = −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q3/5

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2 and x = −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q3/5

True or false?

If the graphs of y = f(x) and y = g(x) intersect at (5, 9), then f(5) = g(5) = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q4/5

QUESTION ID

S19-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q4/5

True or false?

If the graphs of f and g never meet, then f(x) = g(x) has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q5/5

712

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 8.EE.C.8.a

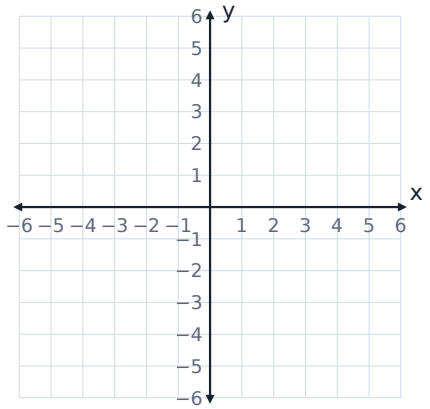
FORWARD 2

Graph the solution set of a system of linear inequalities • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Graph the system on the coordinate plane.

$y \geq x$

$y < 4$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q1/5

714

QUESTION ID

S19-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A solid line y = x and a dashed line y = 4, with the region above or on y = x and below y = 4 shaded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q1/5

Is (1, 3) a solution of the system $y > 2x$ and $y < x + 3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q2/5

715

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 3 > 2(1) and 3 < 1 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q2/5

Which point is a solution of the system $y \leq -x + 4$ and $y > 1$?

A. (1, 2)

B. (4, 3)

C. (3, 2)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q3/5

716

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q3/5

True or false?

The boundary line of $y < 2x + 1$ is drawn dashed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q4/5

717

QUESTION ID

S19-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q4/5

True or false?

The solution set of a system of two inequalities is every point that satisfies at least one of them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q5/5

718

QUESTION ID
S19-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A solution must satisfy both inequalities (the overlap of the two regions).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q5/5

SET 20 • 8.EE.C.8.b • EXPRESSIONS & EQUATIONS

Solve a system of two linear equations algebraically

MAIN	Grade 8 main question	**8.EE.C.8.b**
BACKWARD 1	Evaluate expressions for given values	**6.EE.A.2.c**
BACKWARD 2	Solve equations with the distributive property	**7.EE.B.4.a**
FORWARD 1	Prove that the elimination step keeps the same solutions	**HSA-REI.C.5**
FORWARD 2	Solve a system of a linear equation and a quadratic equation by gr.	**HSA-REI.C.7**

SET 20 • 8.EE.C.8.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system. Show your work.

$y = 3x - 2$

$2x + y = 13$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q1/5

721

QUESTION ID

S20-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

x = 3, y = 7

GRADING NOTE REQUIRED WORK: Any valid method (substitution, elimination, or another), with the steps shown. Example: substitute $y = 3x - 2$ into $2x + y = 13$ to get $5x - 2 = 13$, so $x = 3$ and $y = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system. Show your work.

$x + 2y = 10$

$3x - 2y = 6$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q2/5

722

QUESTION ID

S20-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

x = 4, y = 3

GRADING NOTE REQUIRED WORK: Any valid method, with the steps shown. Example: add the equations to get $4x = 16$, so $x = 4$; then $4 + 2y = 10$, so $y = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q2/5

Solve the system.

$y = x + 4$

$y = -2x + 1$

A. $x = 1, y = 5$

B. $x = 3, y = 7$

C. $x = -1, y = 5$

D. $x = -1, y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q3/5

723

QUESTION ID

S20-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = −1, y = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q3/5

Solve the system.

$4x + 3y = 18$

$2x - y = 4$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q4/5

724

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = 3, y = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q4/5

True or false?

$x = 2, y = -1$ is a solution of both $3x + y = 5$ and $x - y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q5/5

725

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q5/5

SET 20 • 8.EE.C.8.b

BACKWARD 1

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 2x + y when x = 3 and y = 7.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q1/5

727

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q1/5

Evaluate 3x − 2 when x = 4.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q2/5

728

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q2/5

Evaluate x + 2y when x = 4 and y = 3.

A. 9

B. 10

C. 11

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q3/5

729

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q3/5

True or false?

When x = 2, the value of 5x − 1 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q4/5

730

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q4/5

True or false?

When y = 3, the value of 2y + 4 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q5/5

731

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 8.EE.C.8.b

BACKWARD 2

Solve equations with the distributive property • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$5x - 2 = 13$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q1/5

733

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q1/5

Solve for x.

$2x + 3(x - 2) = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q2/5

734

QUESTION ID

S20-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q2/5

Solve for x.

$4x + 2 = -10$

A. $x = 3$

B. $x = -2$

C. $x = -3$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q3/5

735

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q3/5

True or false?

$x = 4$ is the solution of $3x - 5 = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q4/5

736

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q4/5

True or false?

$x = -1$ is the solution of $2x + 9 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q5/5

737

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 8.EE.C.8.b

FORWARD 1

Prove that the elimination step keeps the same solutions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

System A: $x + y = 9$ and $x - y = 3$.

System B: $x + y = 9$ and $2x = 12$ (its second equation is the sum of the two equations of A).

Show that every solution of A is a solution of B, and every solution of B is a solution of A.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q1/5

739

QUESTION ID

S20-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The systems have the same solutions; both have only (6, 3).

GRADING NOTE REQUIRED WORK: Both directions are required. A to B: if (x, y) makes both equations of A true, adding them gives $2x = 12$, so (x, y) solves B. B to A: if (x, y) solves B, subtracting $x + y = 9$ from $2x = 12$ gives $x - y = 3$, so (x, y) solves A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q1/5

System A: $x + 2y = 7$ and $-3x + y = 0$. Replace the second equation by its sum with 3 times the first equation.

Write the new system, and show how to get the original second equation back from it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q2/5

740

QUESTION ID
S20-F1-Q2

TYPE
Short answer

CORRECT ANSWER
New system:
$x + 2y = 7$ and
$7y = 21$. Subtracting
$3(x + 2y) = 21$ from
$7y = 21$ gives
$-3x + y = 0$ again.

GRADING NOTE Grade both: the new system and the reverse step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q2/5

Why does replacing one equation by its sum with a multiple of the other keep the same solutions?

A. Adding equations always gives the solution directly.

B. The new equation always describes the same line as the old one.

C. Multiplying an equation by a number changes its graph.

D. The step can be undone by subtracting the same multiple, so no solution is gained or lost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q3/5

741

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The step can be undone by subtracting the same multiple, so no solution is gained or lost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q3/5

True or false?

The system $x + y = 5$ and $x - y = 1$ has the same solutions as the system $x + y = 5$ and $2x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q4/5

742

QUESTION ID

S20-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q4/5

True or false?

The system x + y = 5 and x − y = 1 has the same solutions as the single equation 2x = 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q5/5

743

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2x = 6 alone is true for (3, y) with any y; the system has only (3, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q5/5

SET 20 • 8.EE.C.8.b

FORWARD 2

Solve a system of a linear equation and a quadratic equation by graphing and algebraically • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system. Give all solutions.

$y = x^2$

$y = 4$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q1/5

745

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

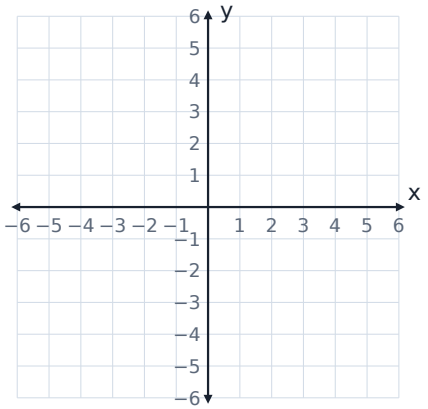
(2, 4) and (−2, 4)

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q1/5

Graph $y = x^2 - 1$ and $y = x + 1$ on the coordinate plane. Mark every intersection point, then check each one in both equations.



Solutions and check: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q2/5

746

QUESTION ID

S20-F2-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The parabola $y = x^2 - 1$ through $(-2, 3)$, $(-1, 0)$, $(0, -1)$, $(1, 0)$, $(2, 3)$, and the line $y = x + 1$, crossing at $(-1, 0)$ and $(2, 3)$

Written: $(2, 3)$ and $(-1, 0)$: $3 = 2^2 - 1 = 2 + 1$ and $0 = (-1)^2 - 1 = -1 + 1$.

GRADING NOTE Grade all three: both graphs, both intersection points, and the algebraic check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q2/5

How many solutions does the system have?

$y = x^2 + 3$

$y = 1$

A. Infinitely many

B. Two

C. One

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q3/5

747

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q3/5

True or false?

The line $y = 2x$ and the parabola $y = x^2$ meet at (0, 0) and (2, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q4/5

748

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q4/5

True or false?

A line and a parabola always meet at exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q5/5

749

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q5/5

SET 21 • 8.EE.C.8.b • EXPRESSIONS & EQUATIONS

Solve simple systems by inspection

MAIN	Grade 8 main question	**8.EE.C.8.b**
BACKWARD 1	Recognize equivalent expressions	**7.EE.A.1**
BACKWARD 2	Decide whether a value makes two equations true	**6.EE.B.5**
FORWARD 1	Use elimination to confirm the number of solutions and solve syst..	**HSA-REI.C.6**
FORWARD 2	Classify lines as parallel, perpendicular, or neither from their equ..	**HSG-GPE.B.5**

SET 21 • 8.EE.C.8.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Without solving, explain why this system has no solution.

3x + 2y = 5

3x + 2y = 6

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q1/5

752

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

The same expression, 3x + 2y, cannot equal both 5 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q1/5

Without solving, how many solutions does this system have? Explain.

$y = 4x - 1$

$y = 4x + 3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q2/5

753

QUESTION ID

S21-M-Q2

TYPE

Short answer

CORRECT ANSWER

None. The lines have the same slope and different y-intercepts, so they are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q2/5

Which system has infinitely many solutions?

A. $y = x$ and $y = -x$

B. $x + y = 4$ and $x + y = 8$

C. $x + y = 4$ and $x - y = 4$

D. $x + y = 4$ and $2x + 2y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q3/5

754

QUESTION ID

S21-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x + y = 4$ and $2x + 2y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q3/5

Without solving, how many solutions does this system have? Explain.

$y = -2x + 5$

$y = 3x + 5$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q4/5

755

QUESTION ID

S21-M-Q4

TYPE

Short answer

CORRECT ANSWER

One. The slopes are different, so the lines cross once (at (0, 5)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q4/5

True or false?

The system $2x - y = 7$ and $6x - 3y = 21$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q5/5

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second equation is 3 times the first, so there are infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q5/5

SET 21 • 8.EE.C.8.b

BACKWARD 1

Recognize equivalent expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Is 2x + 2y equivalent to 2(x + y)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q1/5

758

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. By the distributive property, 2(x + y) = 2x + 2y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q1/5

Write 3(2x + y) without parentheses.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q2/5

759

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6x + 3y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q2/5

Which expression is equivalent to 4x + 6y?

A. 6(4x + y)

B. 4(x + 6y)

C. 2(4x + 6y)

D. 2(2x + 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q3/5

760

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2(2x + 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q3/5

True or false?

$5x - 5y = 5(x - y)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q4/5

761

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q4/5

True or false?

$2(x + 3) = 2x + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q5/5

762

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 8.EE.C.8.b

BACKWARD 2

Decide whether a value makes two equations true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 2 a solution of both x + 3 = 5 and 2x = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q1/5

764

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q1/5

Which of the values 1, 2, and 3 makes both 2x = 6 and x + 1 = 4 true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q2/5

765

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q2/5

Which value makes both x + 2 = 7 and 3x = 15 true?

A. 2

B. 7

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q3/5

766

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q3/5

True or false?

No value of x makes both x + 1 = 3 and x + 1 = 4 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q4/5

767

QUESTION ID

S21-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$x = 4$ makes both $x - 1 = 3$ and $2x = 6$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q5/5

768

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 8.EE.C.8.b

FORWARD 1

Use elimination to confirm the number of solutions and solve systems with letter coefficients • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Add 2 times the first equation to the second equation. What equation results, and what does it tell you about the system?

$x - 3y = 4$

$-2x + 6y = 1$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q1/5

770

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0 = 9, which is false, so the system has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q1/5

Solve the system for x and y in terms of a.

$x + y = 3a$

$x - y = a$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q2/5

771

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2a and y = a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q2/5

Elimination on a system of two linear equations gives 0 = 0.

What does this tell you?

A. There is exactly one solution.

B. The only solution is (0, 0).

C. The equations describe the same line, so there are infinitely many solutions.

D. There is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q3/5

772

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The equations describe the same line, so there are infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q3/5

True or false?

The system $x + y = 1.5$ and $2x - y = 0$ has the solution $x = 0.5$, $y = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q4/5

773

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q4/5

True or false?

If elimination gives 0 = 5, the system has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q5/5

774

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0 = 5 is false, so there is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 8.EE.C.8.b

FORWARD 2

Classify lines as parallel, perpendicular, or neither from their equations • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Are the lines y = 2x + 1 and 4x − 2y = 6 parallel, perpendicular, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q1/5

776

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Parallel; both have slope 2 (and different y-intercepts).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q1/5

Are the lines 3x + y = 2 and x − 3y = 9 parallel, perpendicular, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q2/5

777

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Perpendicular; the slopes are −3 and 1/3, and their product is −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q2/5

One line passes through (0, 0) and (2, 4). Another passes through (1, 1) and (3, 5).

How are the lines related?

A. Perpendicular

B. The same line

C. Parallel

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q3/5

778

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q3/5

True or false?

Lines with slopes $\frac{2}{3}$ and $-\frac{3}{2}$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q4/5

779

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q4/5

True or false?

The lines y = x and y = −x are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q5/5

780

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q5/5

SET 22 • 8.EE.C.8.c • EXPRESSIONS & EQUATIONS

Write and solve systems of linear equations for real-world problems

MAIN	Grade 8 main question	**8.EE.C.8.c**
BACKWARD 1	Solve word problems with one unknown	**7.EE.B.4.a**
BACKWARD 2	Use variables to represent quantities in a situation	**6.EE.B.6**
FORWARD 1	Create cost, revenue, and profit equations in two variables and gr.	**HSA-CED.A.2**
FORWARD 2	Represent constraints with a system of inequalities	**HSA-CED.A.3**

SET 22 • 8.EE.C.8.c

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.c

A school sells 200 tickets. Adult tickets cost $6 and student tickets cost $4. The sales total $940.

Write a system and solve it to find how many of each ticket were sold.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q1/5

783

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

The system a + s = 200 and 6a + 4s = 940, and the answer 70 adult and 130 student tickets

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Plan A costs $15 plus $0.05 per minute. Plan B costs $0.10 per minute.

Write a system and solve it to find when the plans cost the same.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q2/5

784

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

The system c = 15 + 0.05m and c = 0.10m, and the answer 300 minutes (both cost $30)

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q2/5

Jo has 25 coins, all dimes and quarters, worth $4.00 in all.

Which system models this?

A. $d - q = 25$ and $0.10d + 0.25q = 4$

B. $d + q = 25$ and $0.10d + 0.25q = 4$

C. $d + q = 4$ and $0.10d + 0.25q = 25$

D. $d + q = 25$ and $0.25d + 0.10q = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q3/5

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $d + q = 25$ and $0.10d + 0.25q = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q3/5

Jo has 25 coins, all dimes and quarters, worth $4.00 in all: d + q = 25 and 0.10d + 0.25q = 4.

How many of each coin does Jo have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q4/5

786

QUESTION ID

S22-M-Q4

TYPE

Short answer

CORRECT ANSWER

15 dimes and 10 quarters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q4/5

True or false?

Two numbers have a sum of 40 and a difference of 12. The larger number is 26.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q5/5

787

QUESTION ID

S22-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q5/5

SET 22 • 8.EE.C.8.c

BACKWARD 1

Solve word problems with one unknown • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A taxi charges $3 plus $2 per mile. A ride costs $19.

How many miles long was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q1/5

789

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q1/5

Concert tickets cost $9 each, plus a $5 fee for the whole order. An order costs $50.

How many tickets were bought?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q2/5

790

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q2/5

The sum of a number and 3 times the number is 48.

What is the number?

A. 16

B. 45

C. 12

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q3/5

791

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q3/5

True or false?

If 2x + 7 = 25, then x = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q4/5

792

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q4/5

True or false?

Items cost $4 each, and shipping is $6. A $30 order has 7 items.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q5/5

793

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4n + 6 = 30 gives n = 6 items.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 8.EE.C.8.c

BACKWARD 2

Use variables to represent quantities in a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Write an expression for the value, in dollars, of d dimes.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

795

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.10d

GRADING NOTE 0.1d is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

Adult tickets cost $6.

Write an expression for the cost of a adult tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

796

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

There are 25 coins: d dimes and the rest quarters.

Which expression gives the number of quarters?

A. d − 25

B. 25d

C. 25 − d

D. d + 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

797

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 25 − d

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

True or false?

q quarters are worth 0.25q dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

798

QUESTION ID

S22-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

There are s students and 3 more teachers than students. There are 3s teachers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

799

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There are s + 3 teachers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 8.EE.C.8.c

FORWARD 1

Create cost, revenue, and profit equations in two variables and graph them • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2